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Liaw

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(54) **METAL GATE STACKS AND METHODS OF FABRICATING THE SAME IN MULTI-GATE FIELD-EFFECT TRANSISTORS**

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**,
Hsin-Chu (TW)

(72) Inventor: **Jhon Jhy Liaw**, Hsinchu County (TW)

(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.**, Hsinchu (TW)

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Related U.S. Application Data

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H10D 30/01 (2025.01)
H10D 30/62 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)

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CPC H01L 27/0886; H01L 21/823418; H01L 21/823431; H01L 29/6681; H01L 29/7851; H01L 29/7855; H10D 30/6211; H10D 30/6215; H10D 30/0243; H10D 84/834; H10D 84/013; H10D 84/0158; H10D 84/038

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,379,221 B1 6/2016 He et al.
10,879,308 B1 * 12/2020 Ando H10B 63/845
2019/0131394 A1 * 5/2019 Reznicek H01L 21/30604
2019/0378911 A1 12/2019 Lee et al.
2020/0013870 A1 * 1/2020 Ha H01L 29/42392

(Continued)

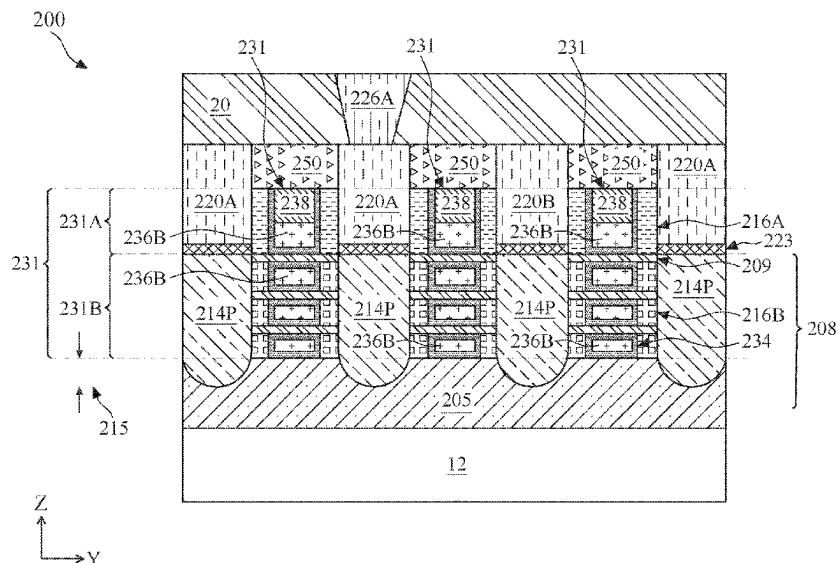
Primary Examiner — Christine A Enad

(74) *Attorney, Agent, or Firm* — HAYNES AND BOONE, LLP

(57) **ABSTRACT**

A semiconductor structure includes a substrate, a semiconductor fin protruding from the substrate, where the semiconductor fin includes semiconductor layers stacked in a vertical direction, a gate stack engaging with channel regions of the semiconductor fin, and source/drain (S/D) features disposed adjacent to the gate stack in S/D regions of the semiconductor fin. In the present embodiments, the gate stack includes a first portion disposed over the semiconductor layers and a second portion disposed between the semiconductor layers, where the first portion includes a work-function metal (WFM) layer and a metal fill layer disposed over the WFM layer and the second portion includes the WFM layer but is free of the metal fill layer.

20 Claims, 26 Drawing Sheets



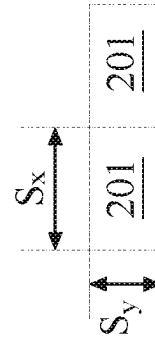
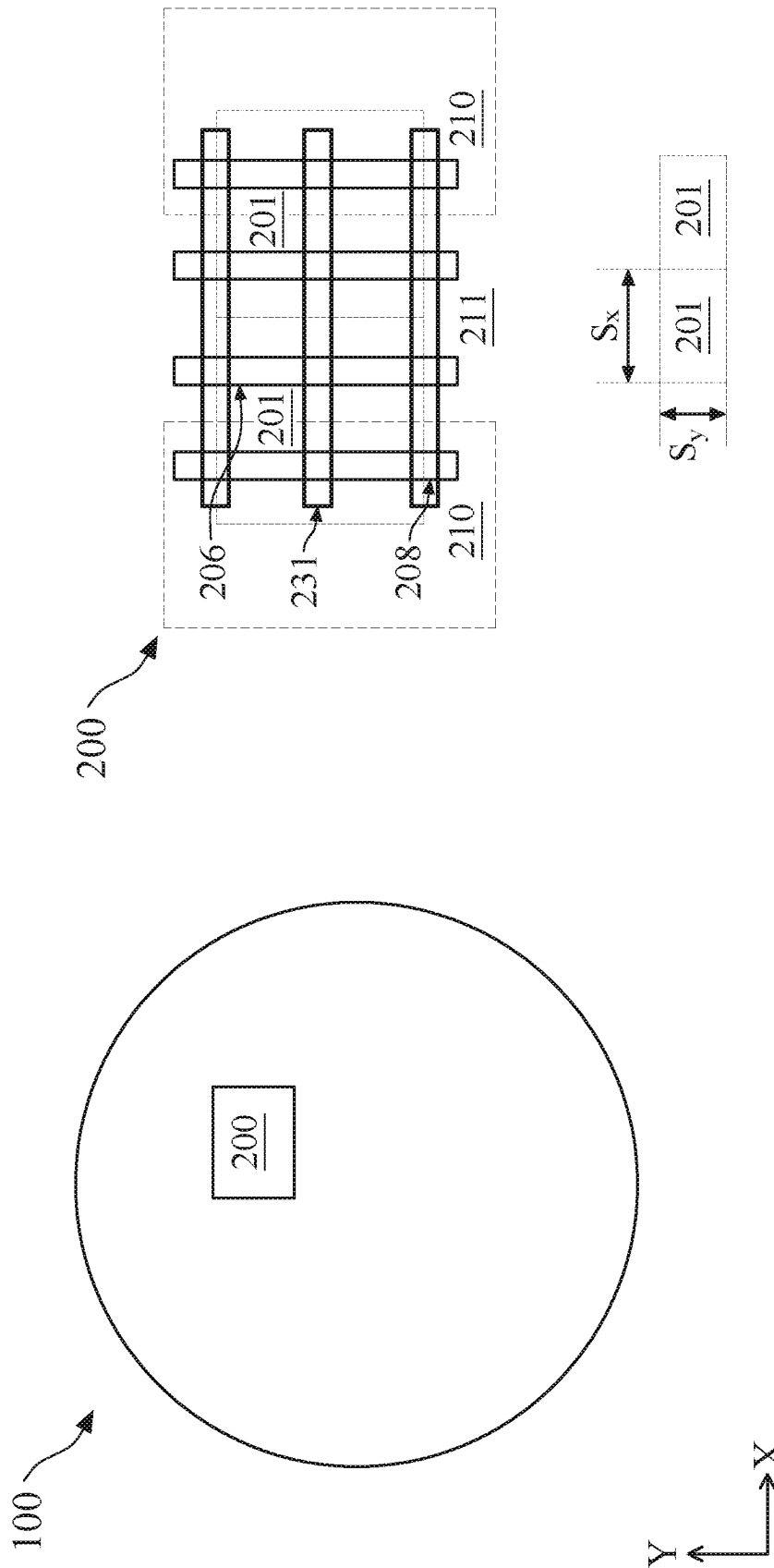
(56)

References Cited

U.S. PATENT DOCUMENTS

2020/0127054	A1 *	4/2020	Ando	H10N 70/021
2021/0305380	A1 *	9/2021	Bouche	H01L 29/78696
2022/0059533	A1 *	2/2022	Bae	H01L 21/823821

* cited by examiner



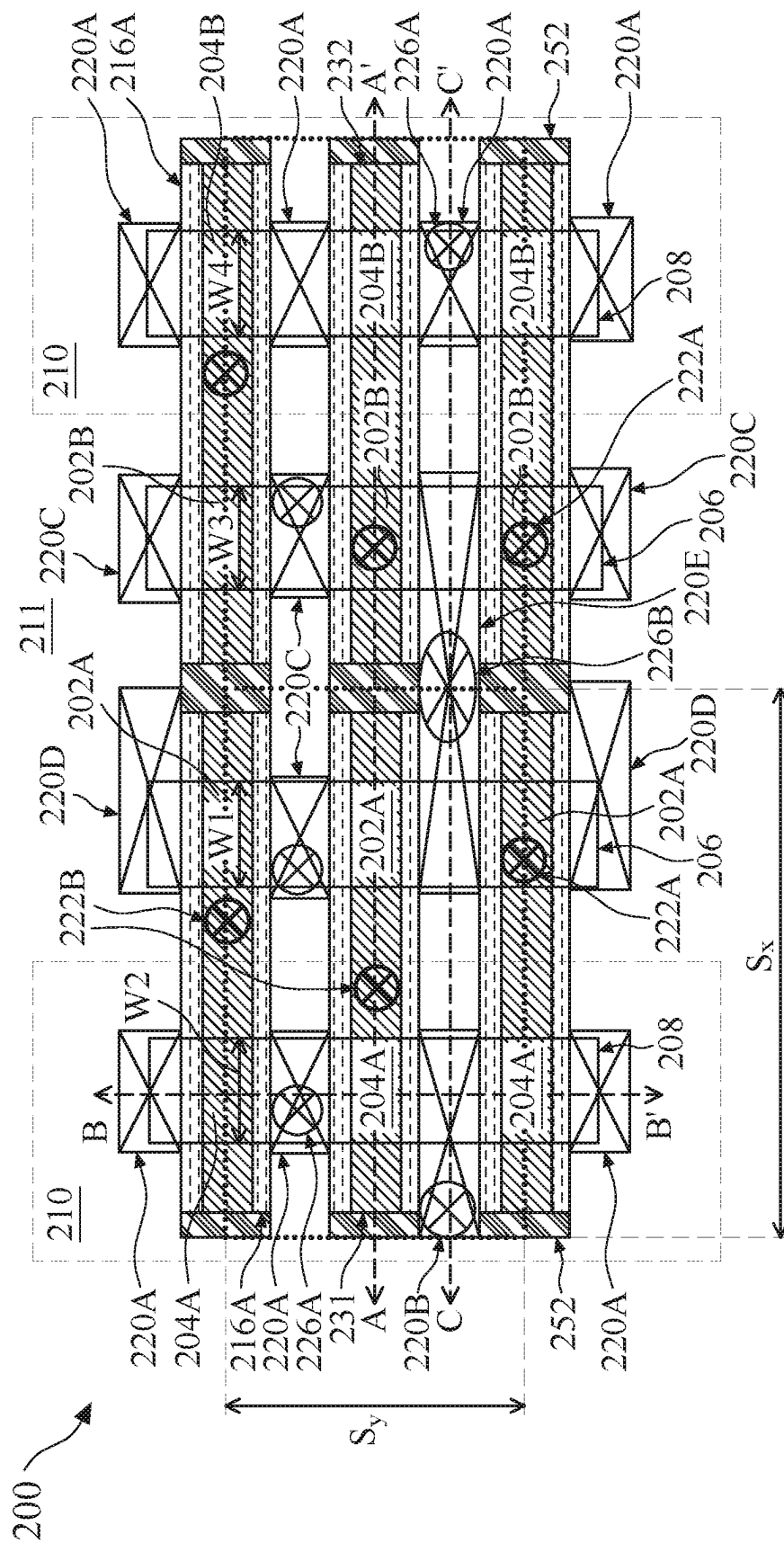
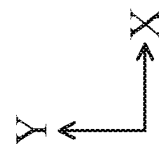
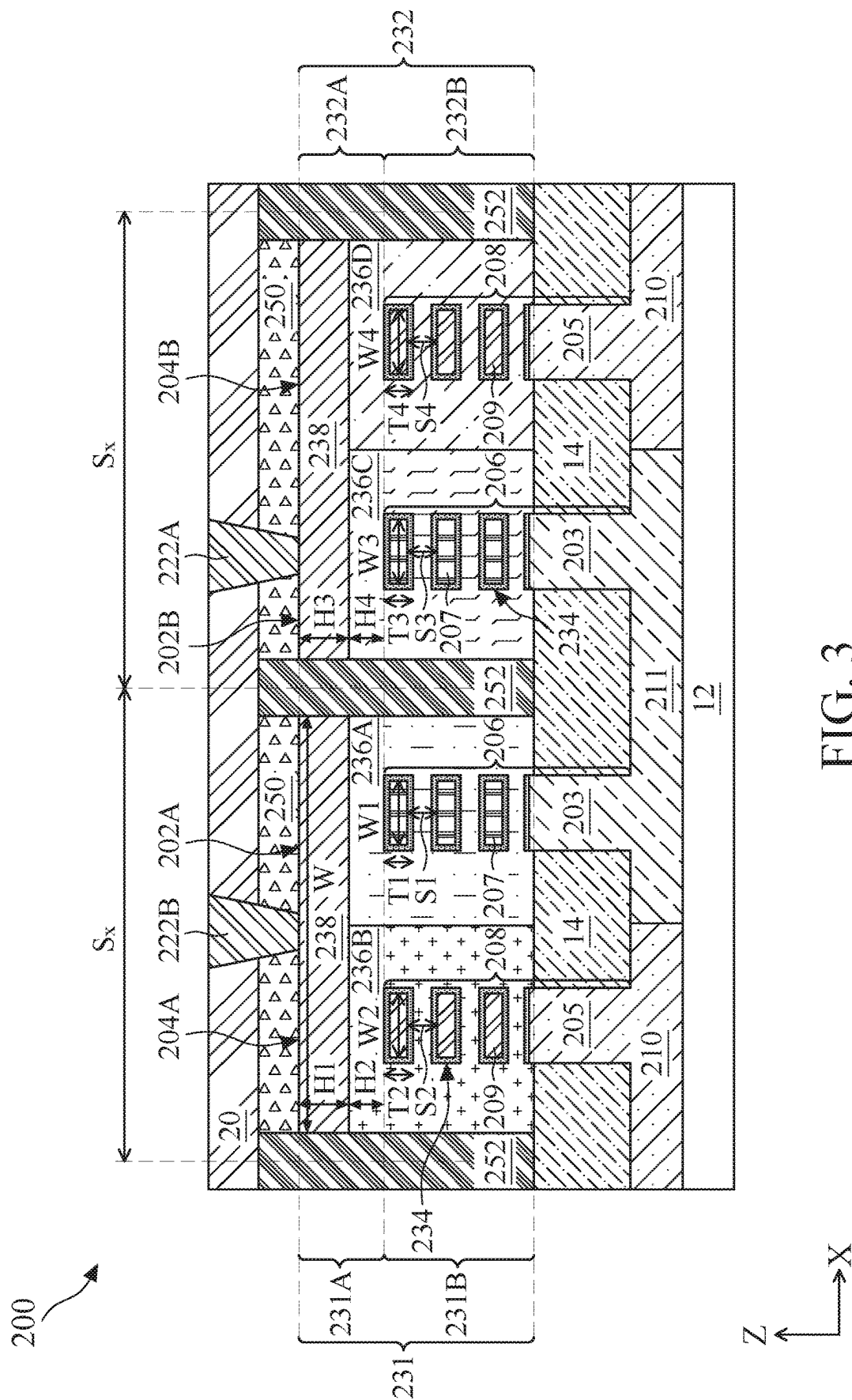
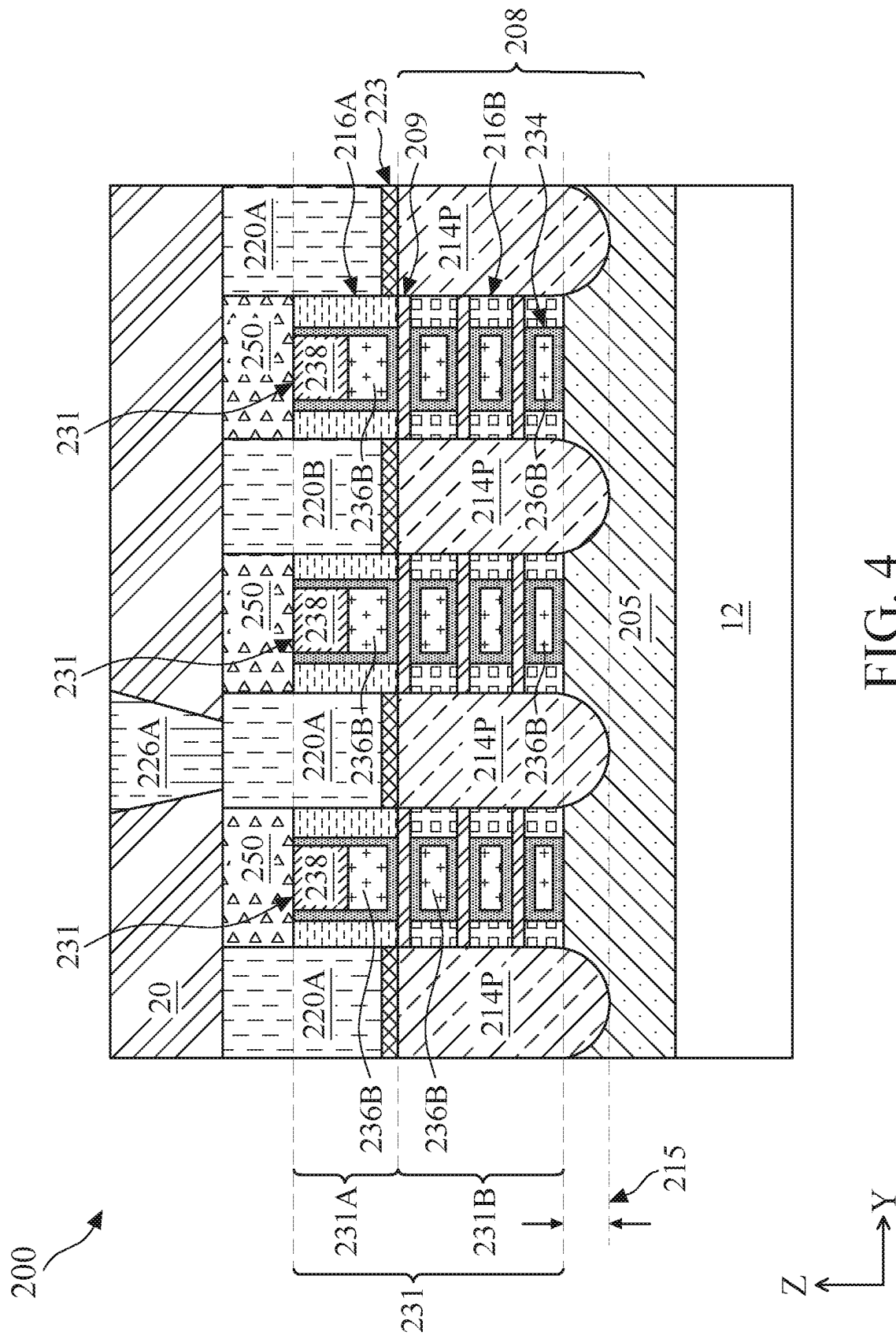
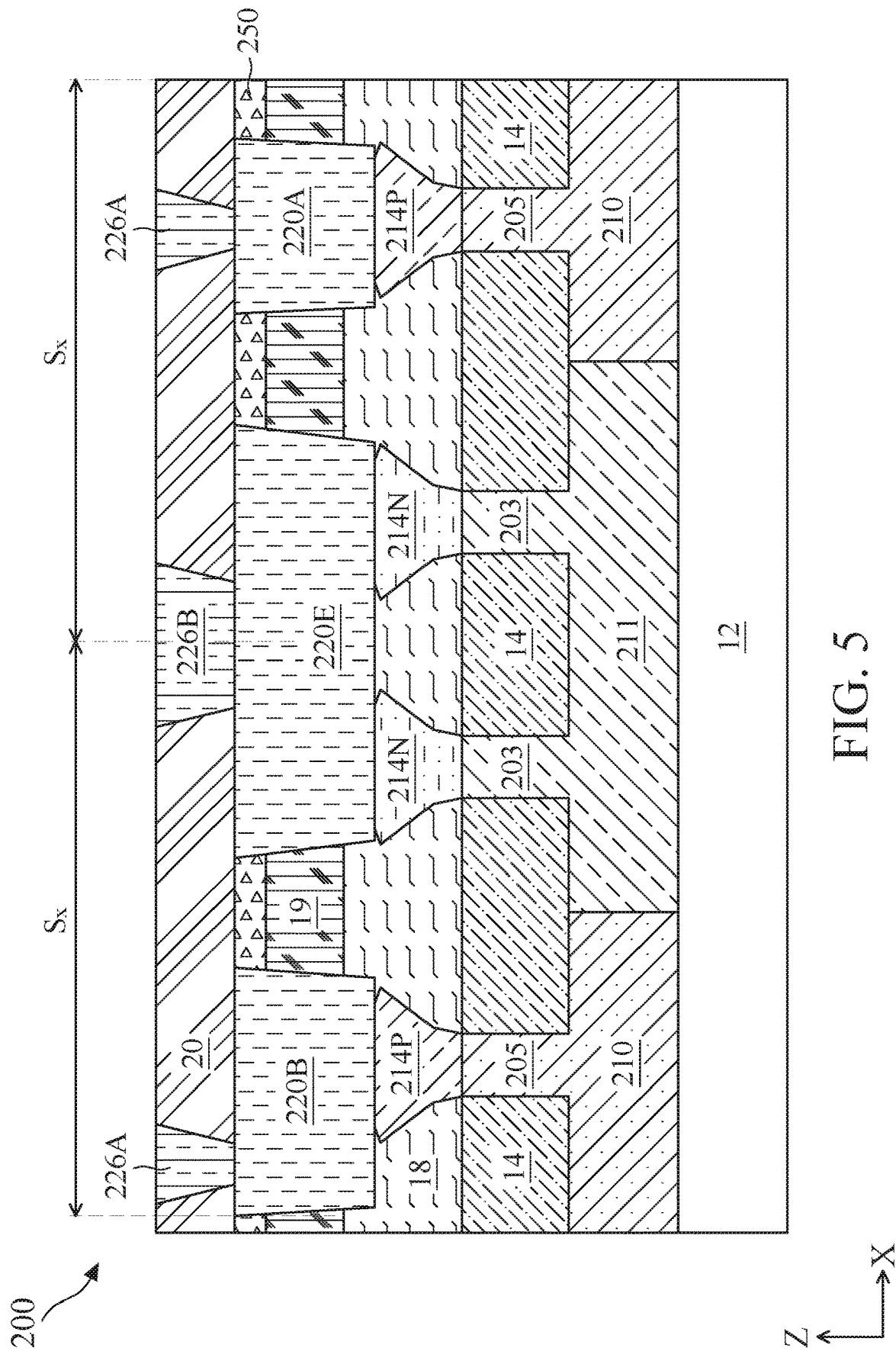


FIG. 2









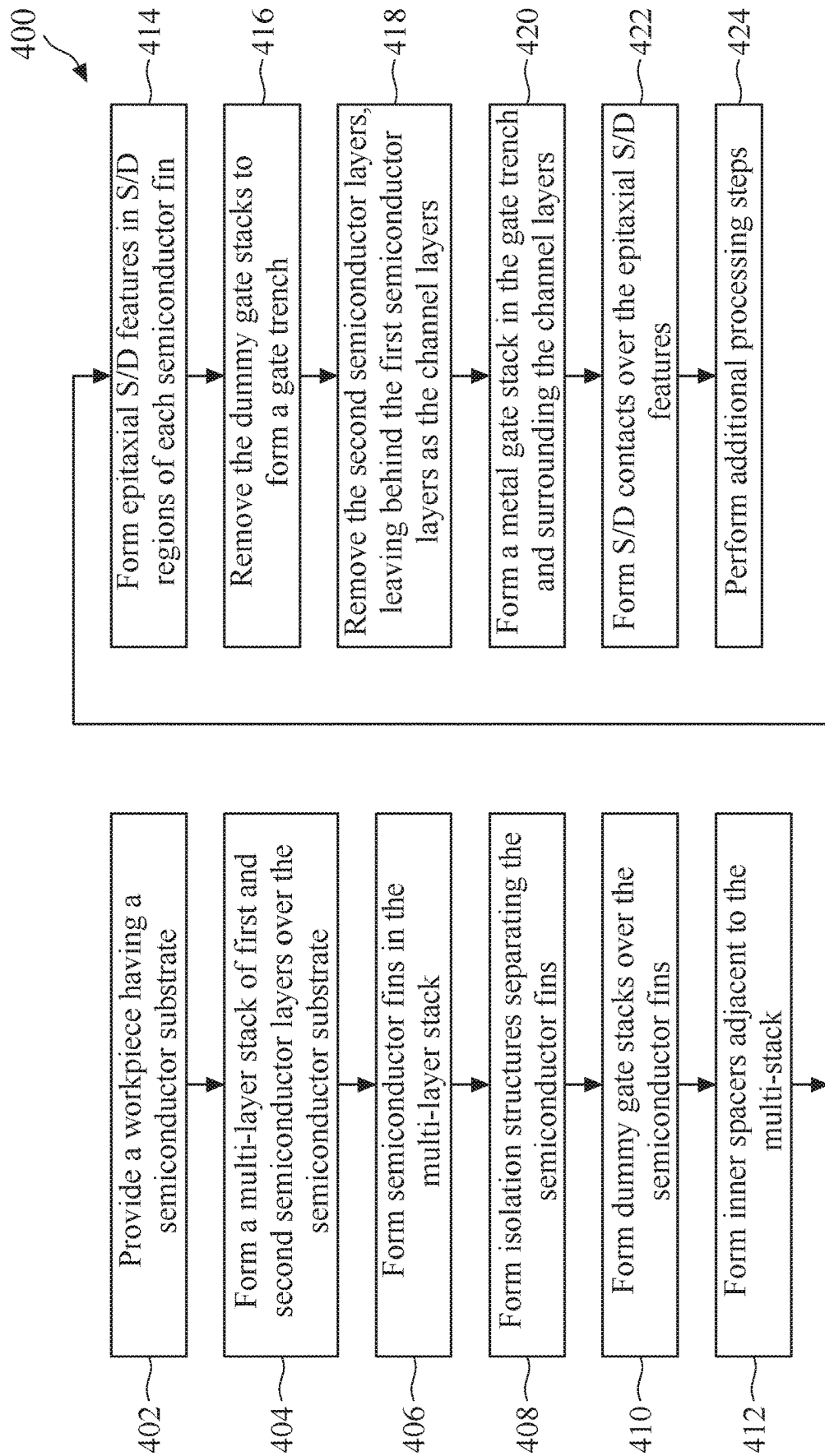


FIG. 6A

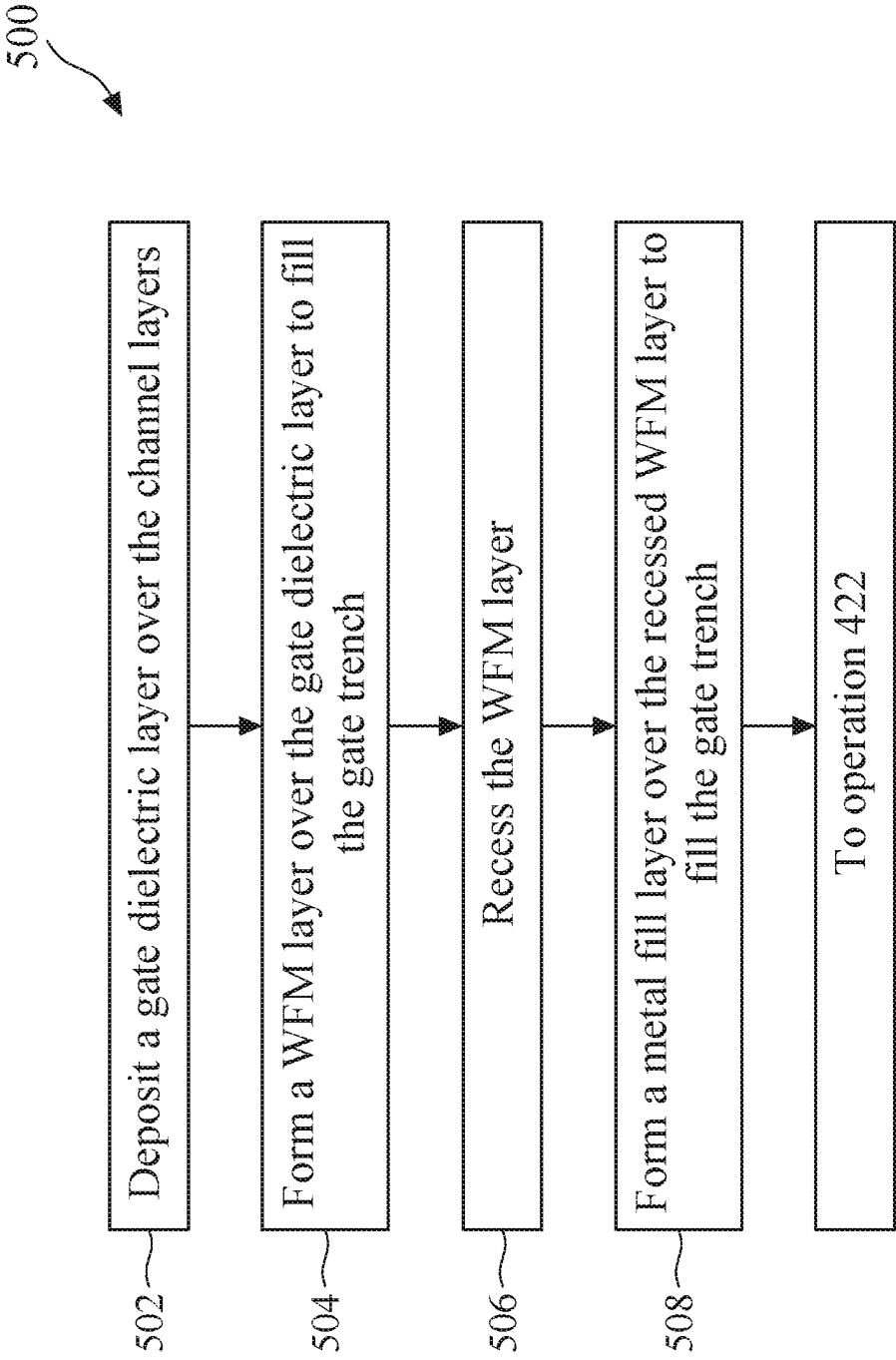


FIG. 6B

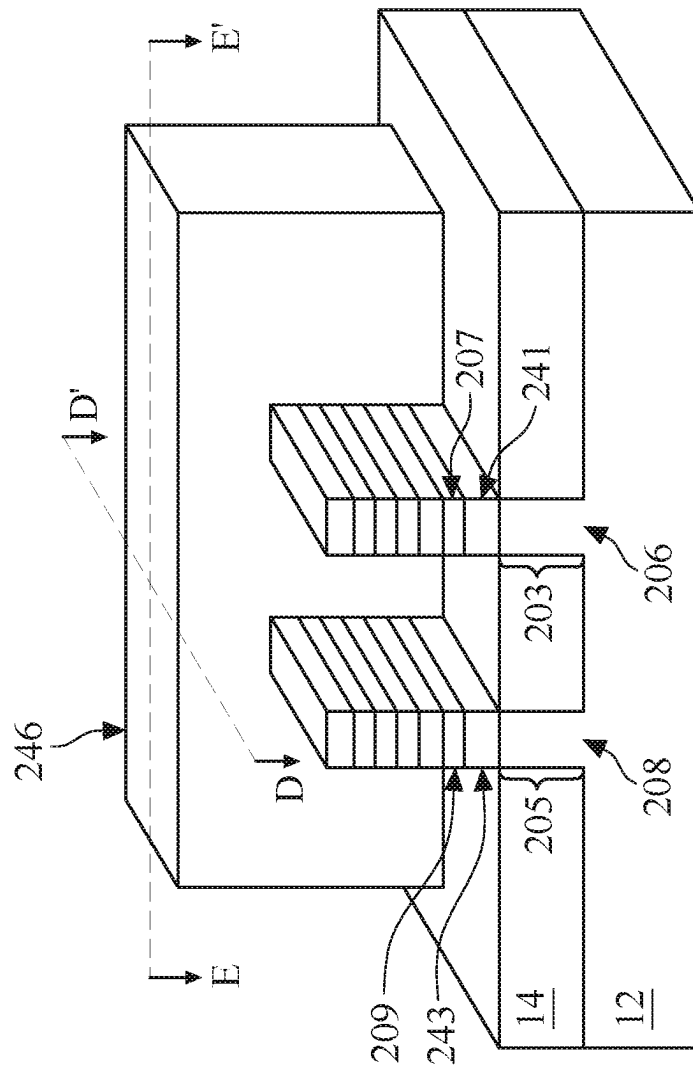
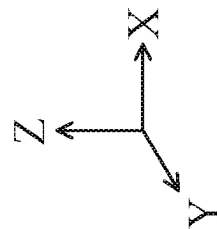


FIG. 7



200

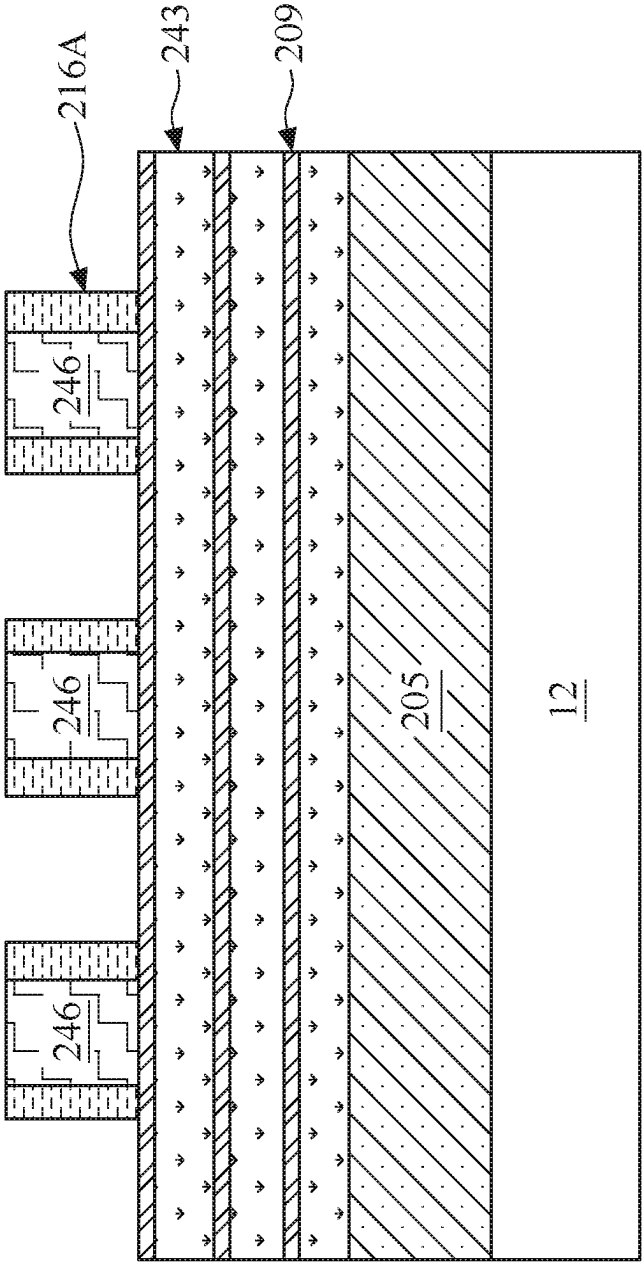


FIG. 8A

200

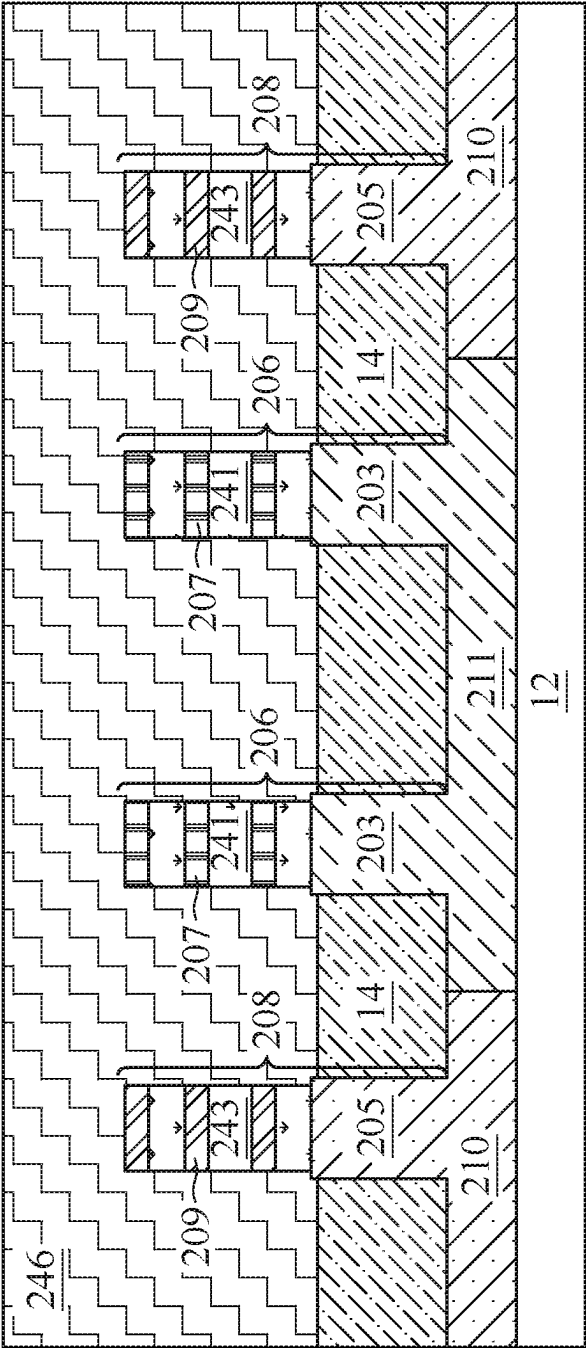


FIG. 8B

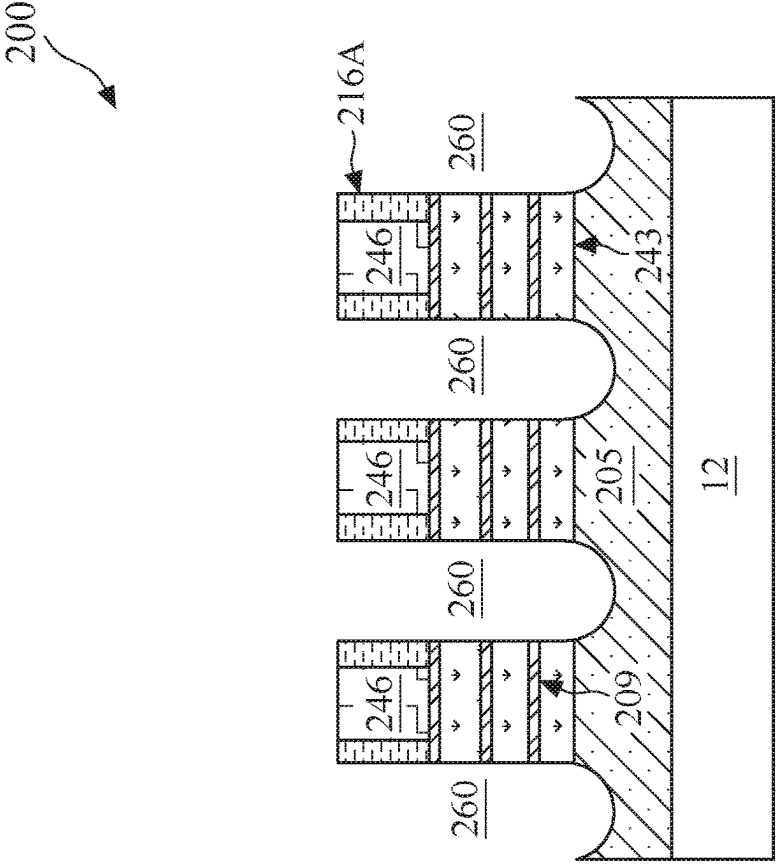
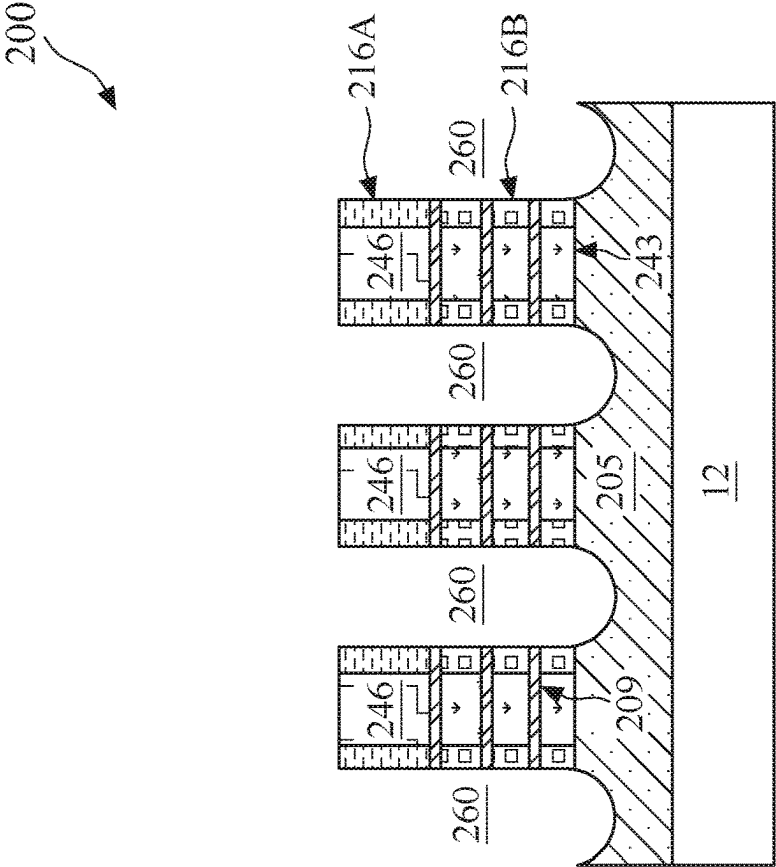
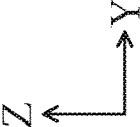


FIG. 9

FIG. 10



200

200

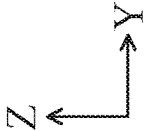
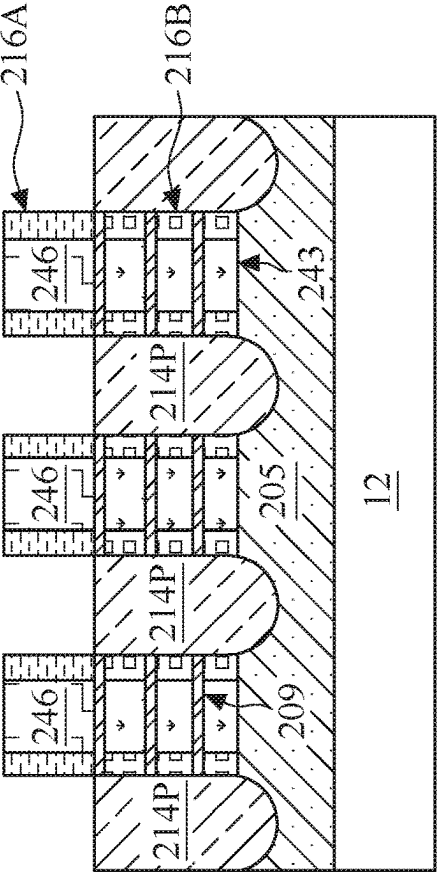
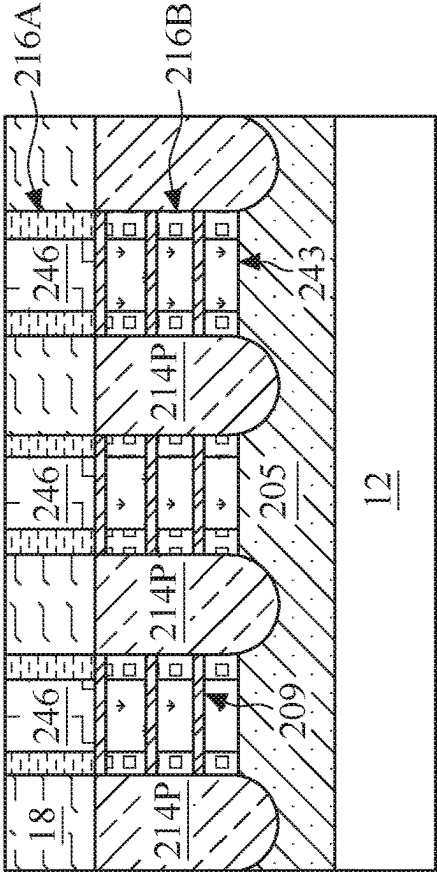


FIG. 12

FIG. 11

200

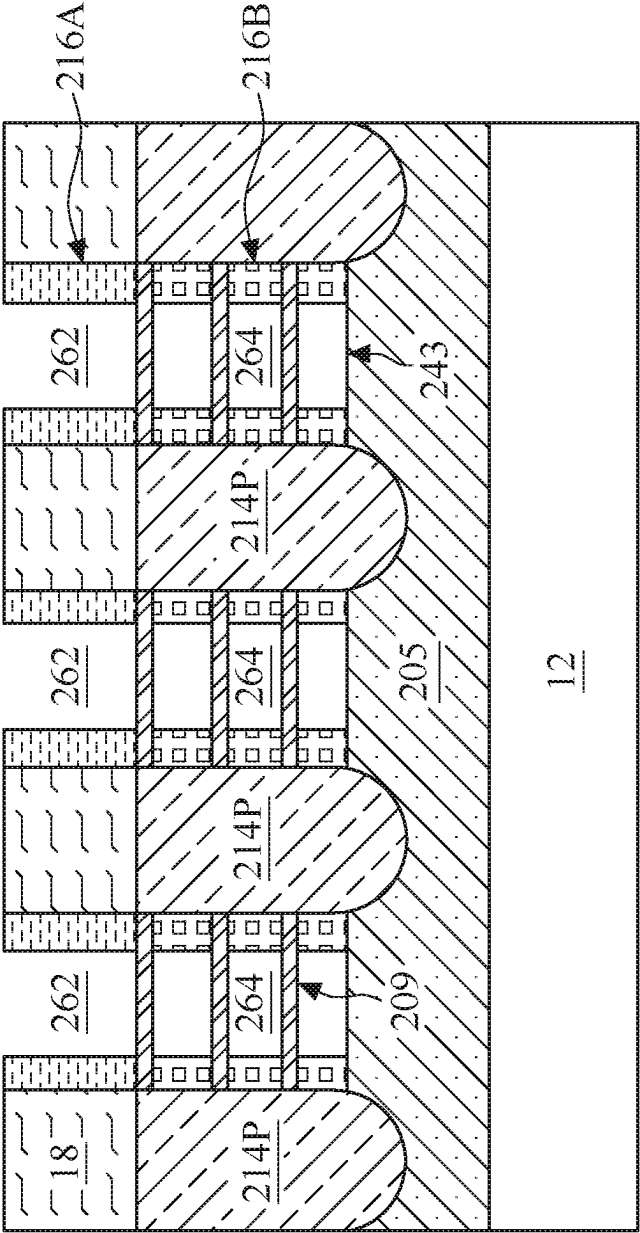
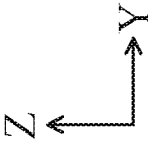


FIG. 13A



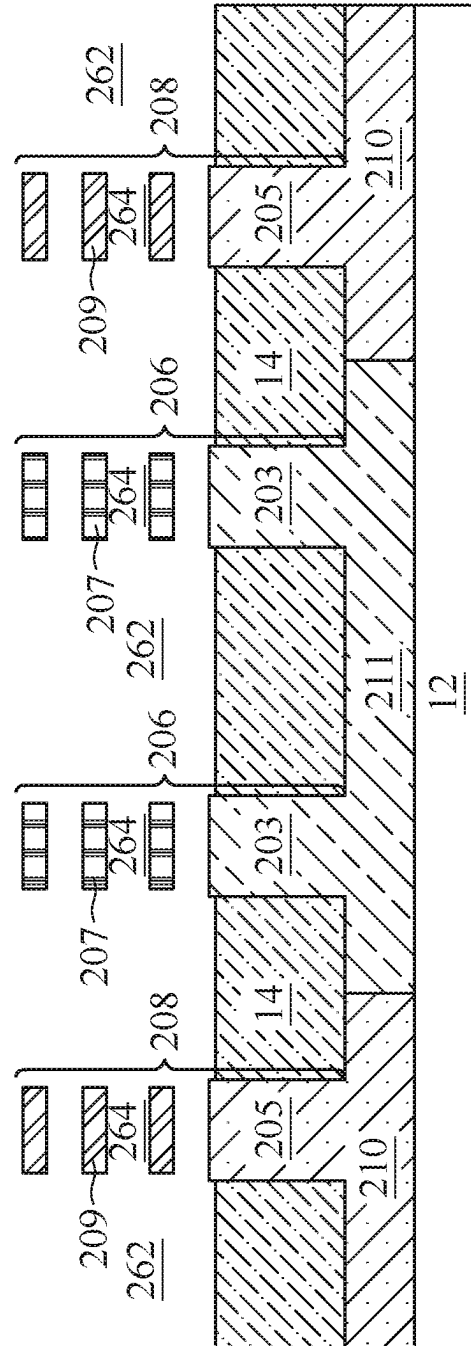
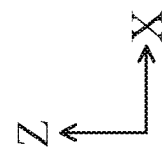


FIG. 13B



200

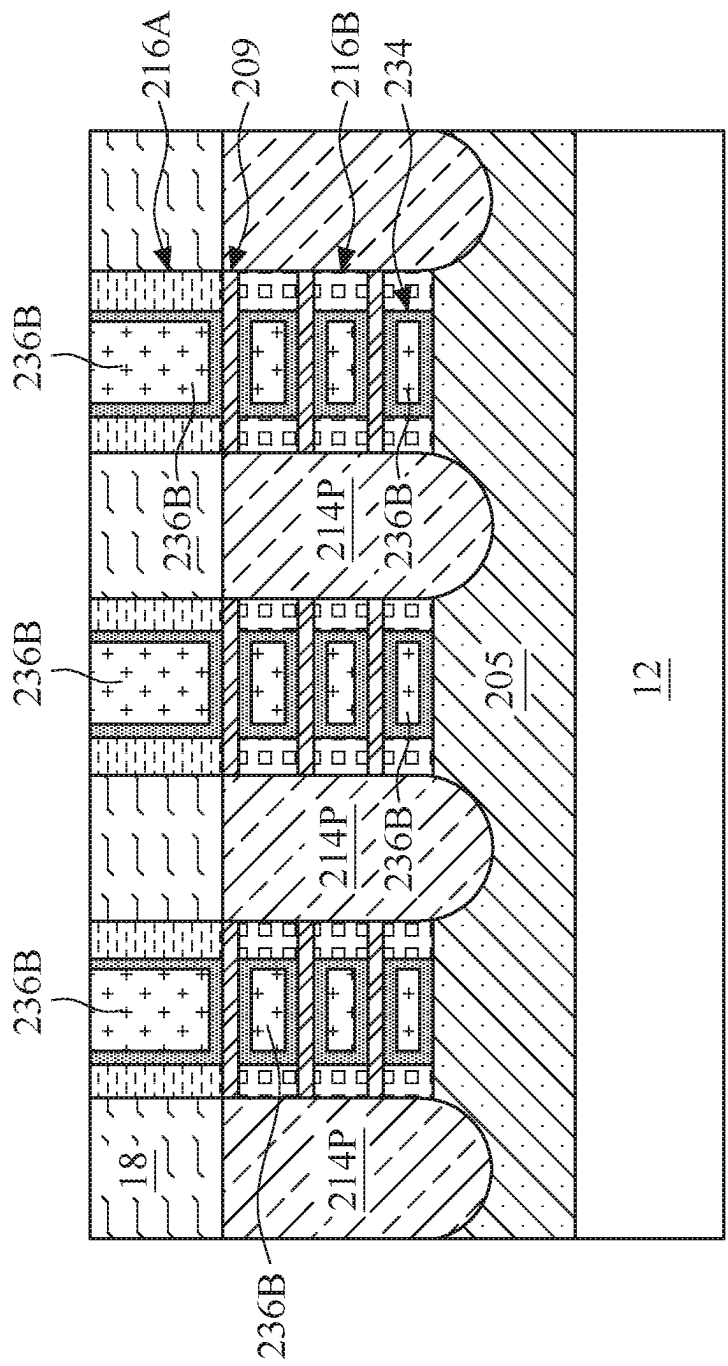


FIG. 14A

200

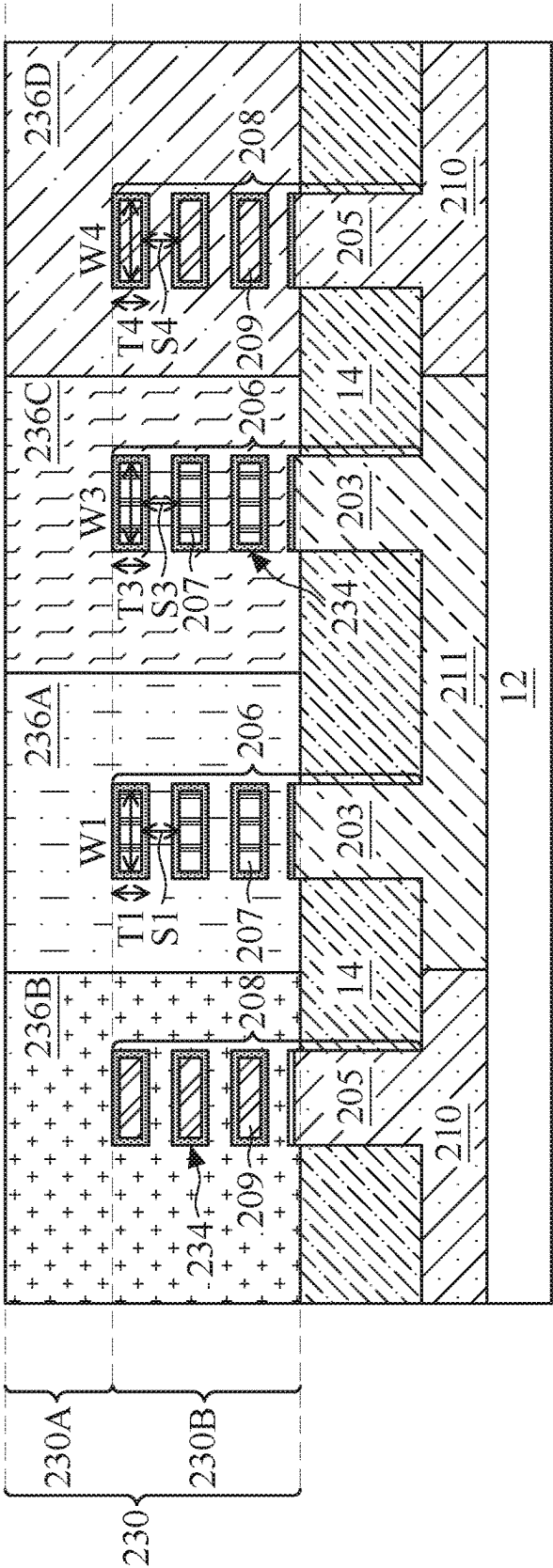


FIG. 14B

200

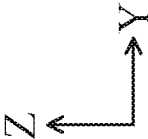
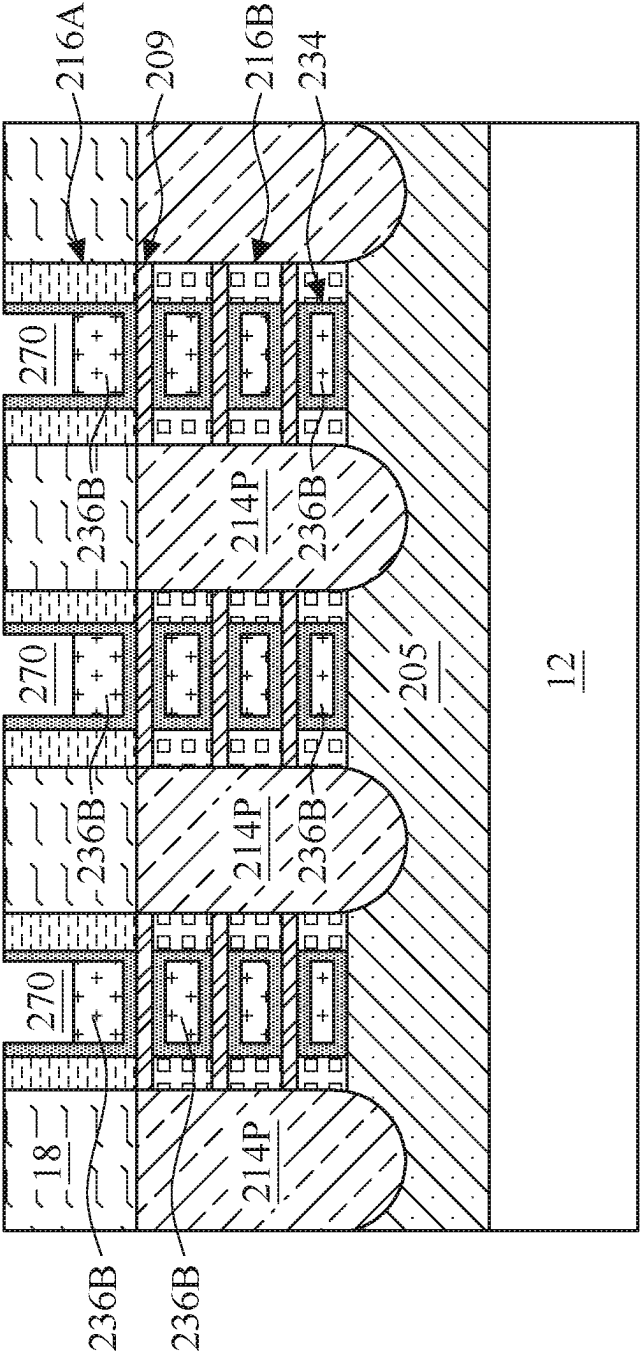
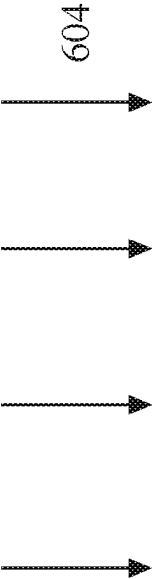


FIG. 15A

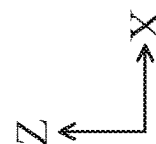
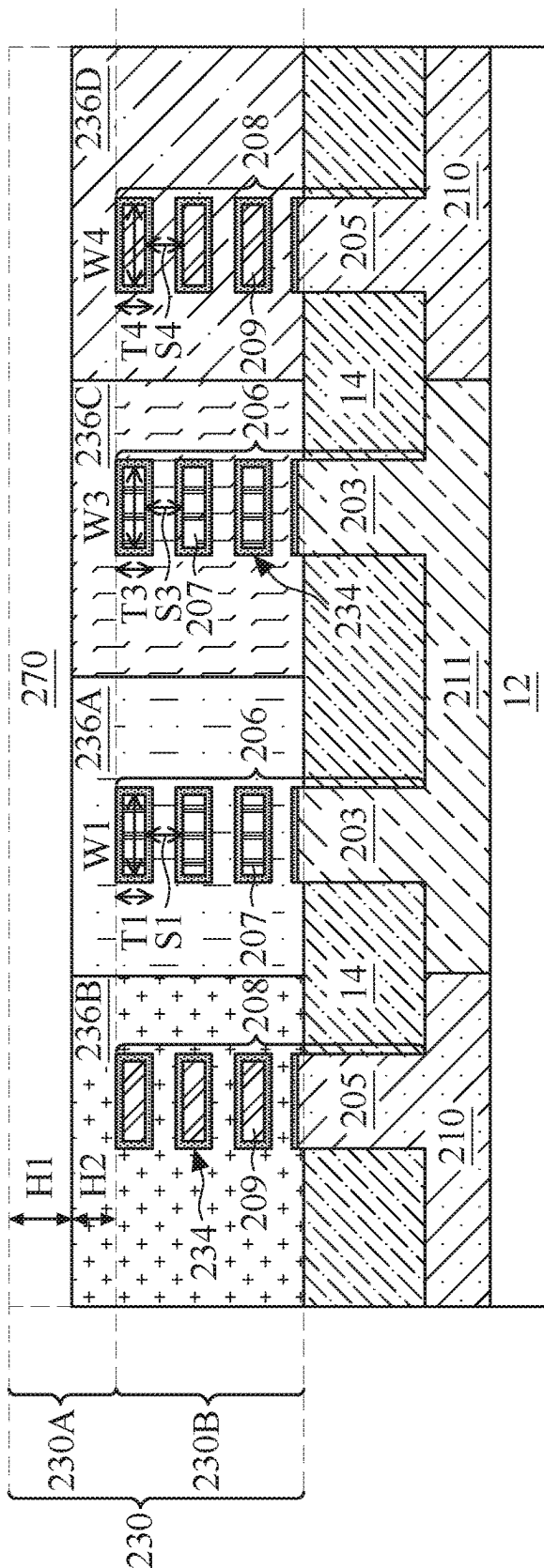
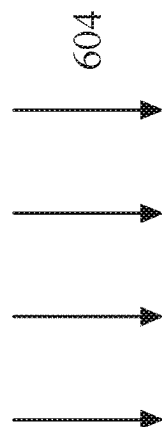
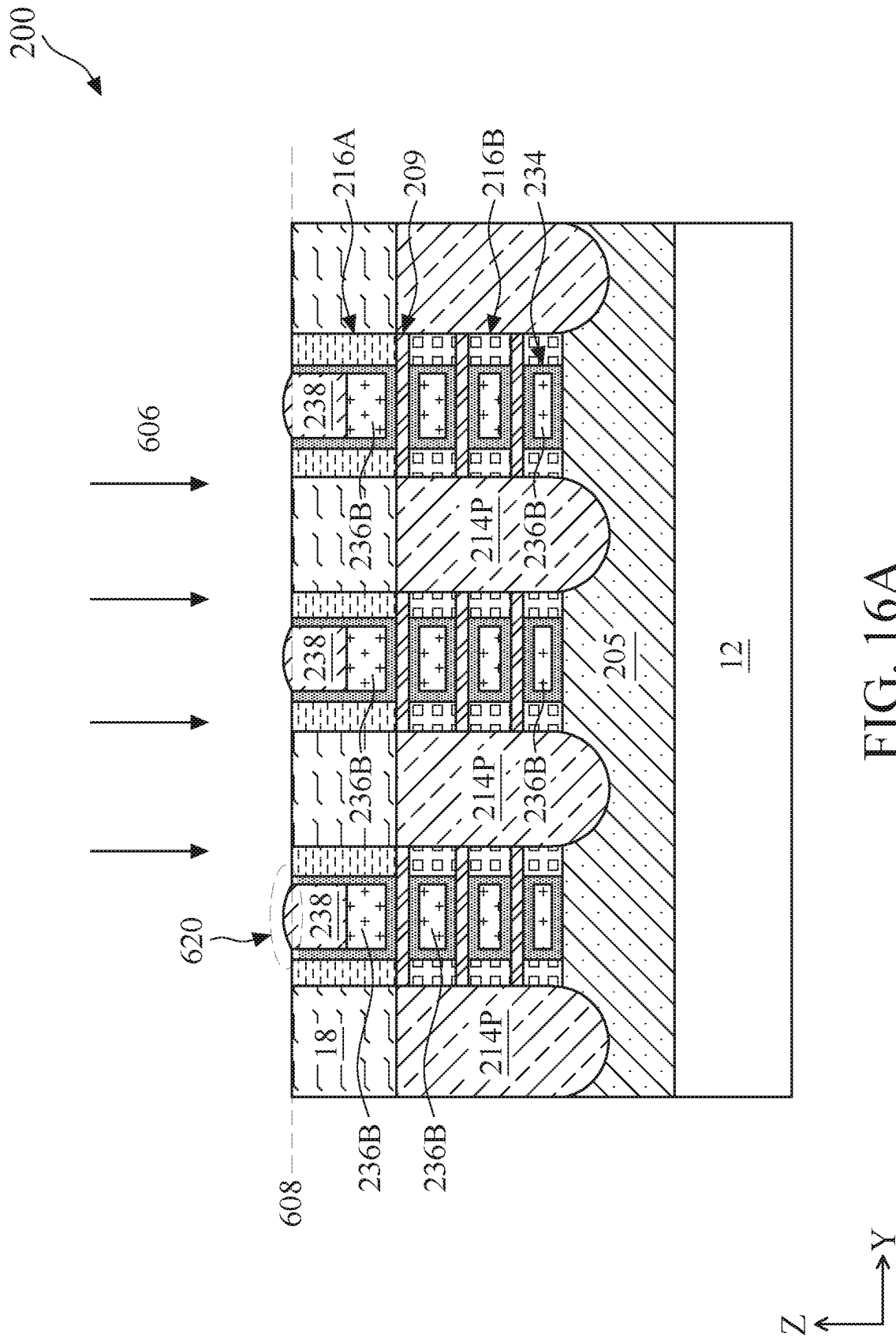
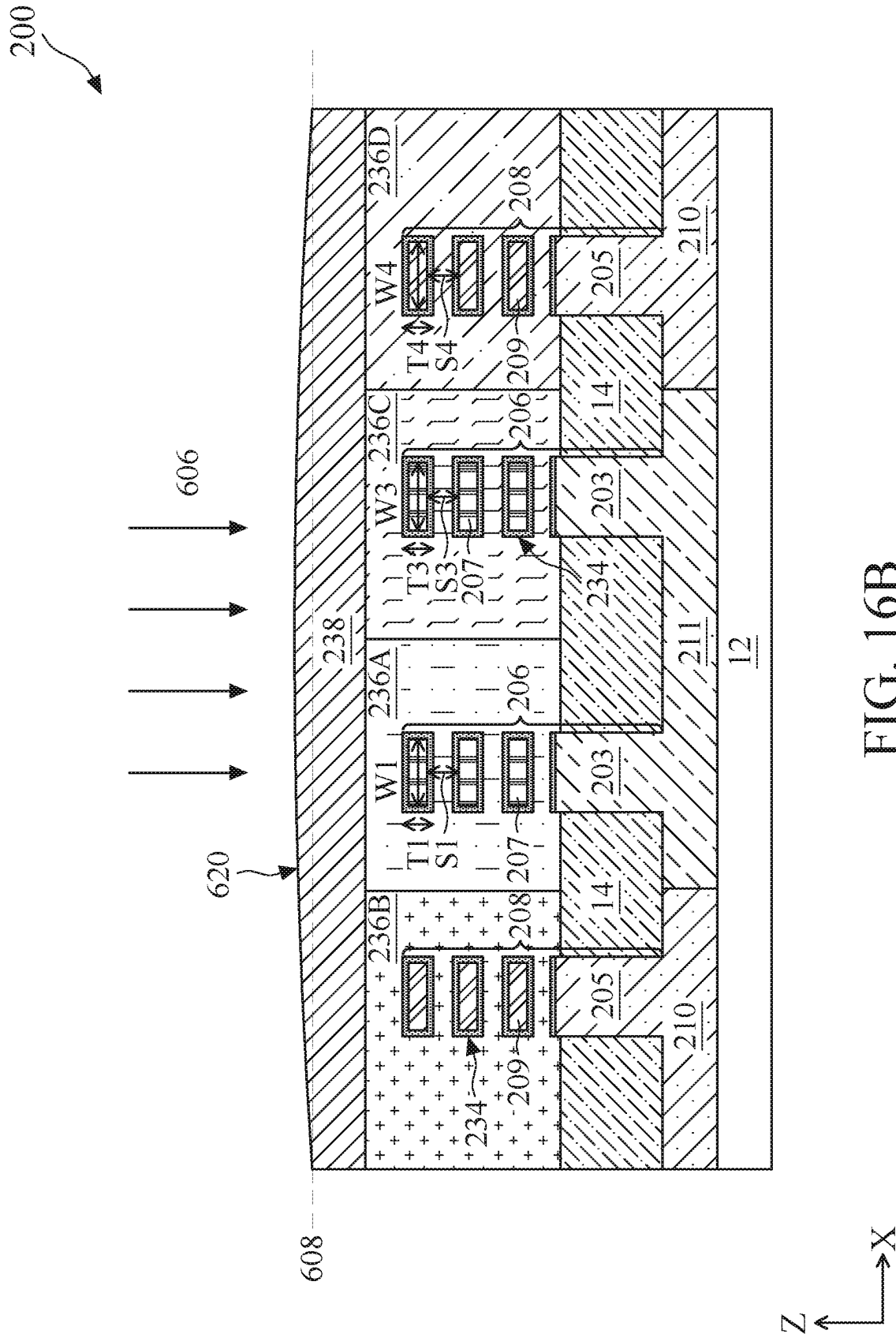


FIG. 15B





200

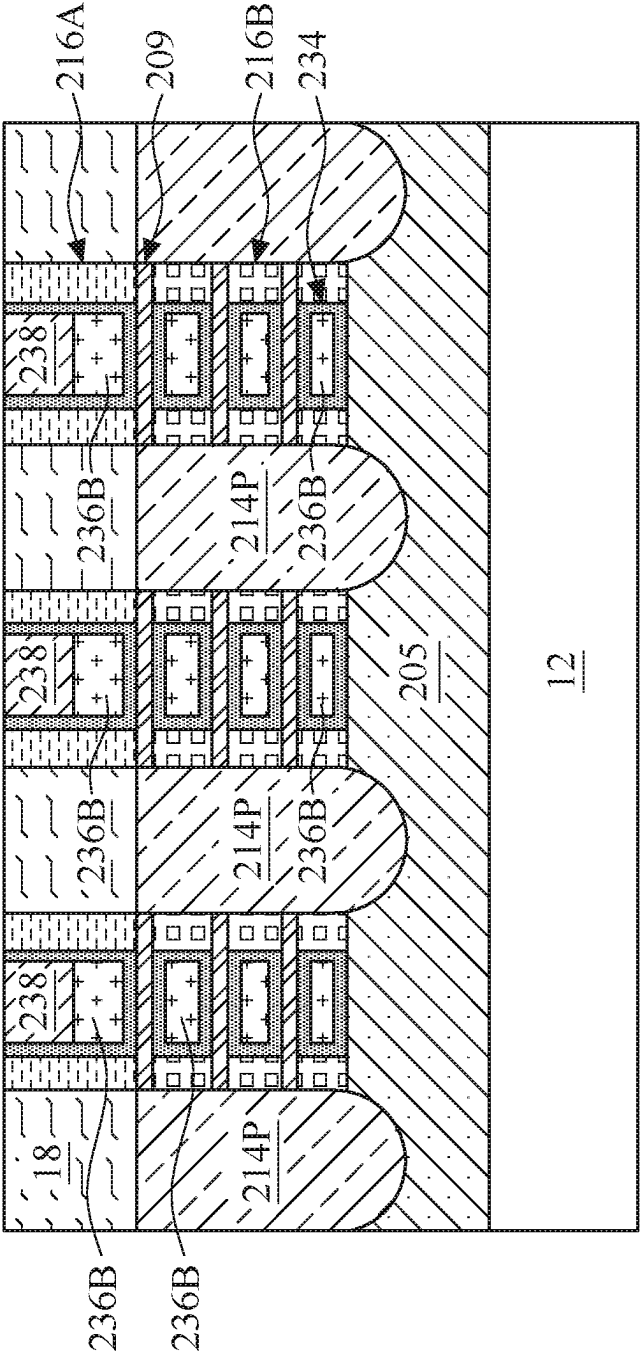


FIG. 17A

200

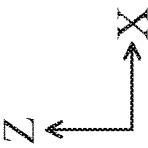
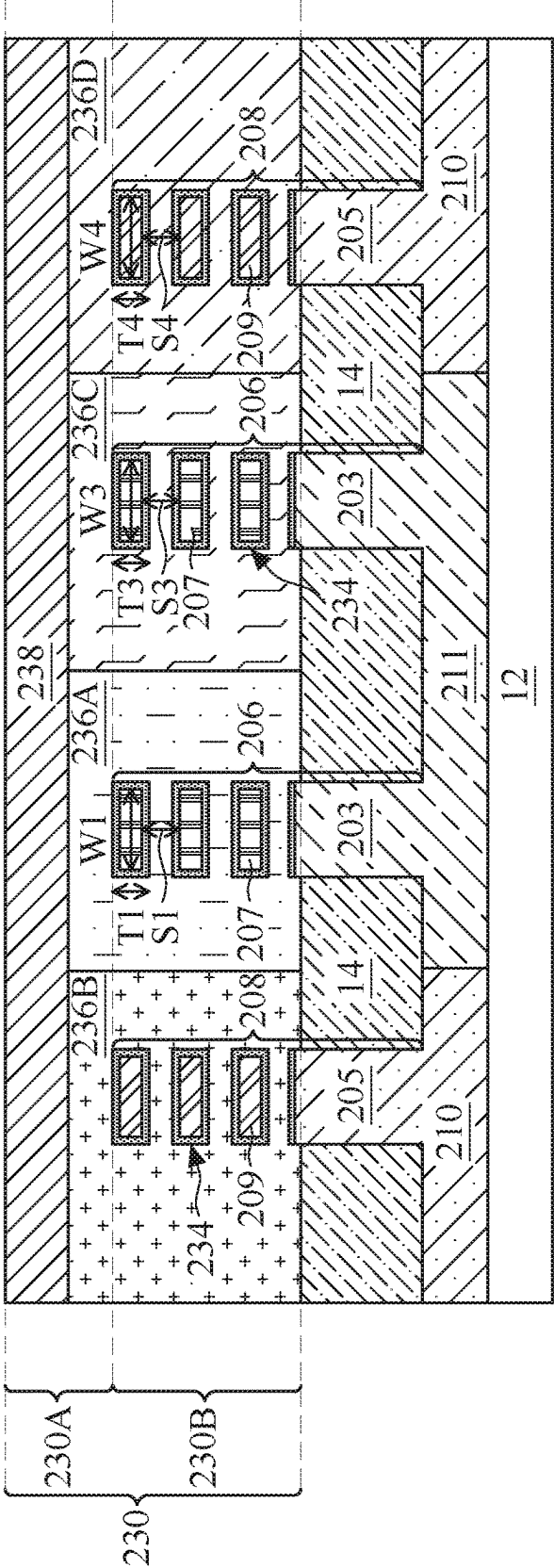


FIG. 17B

200

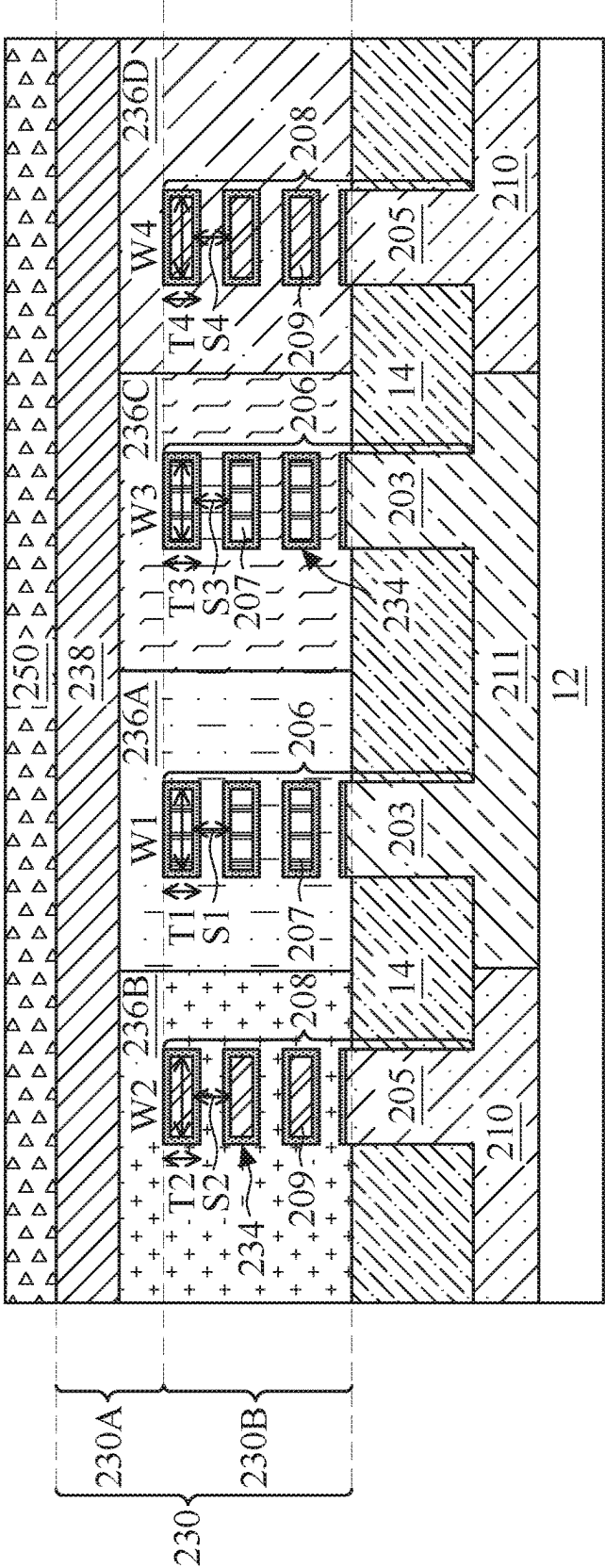


FIG. 18B

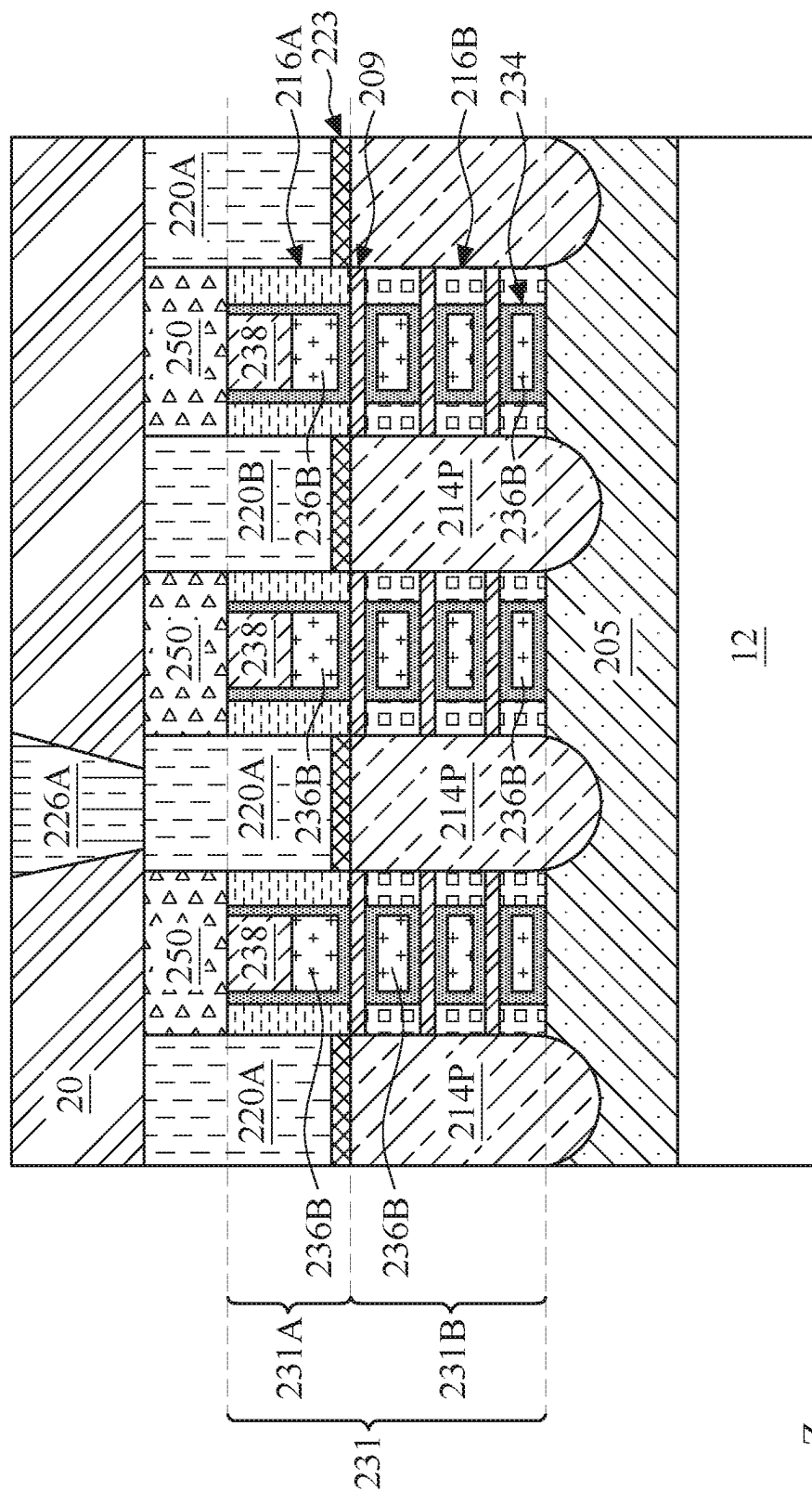


FIG. 19A

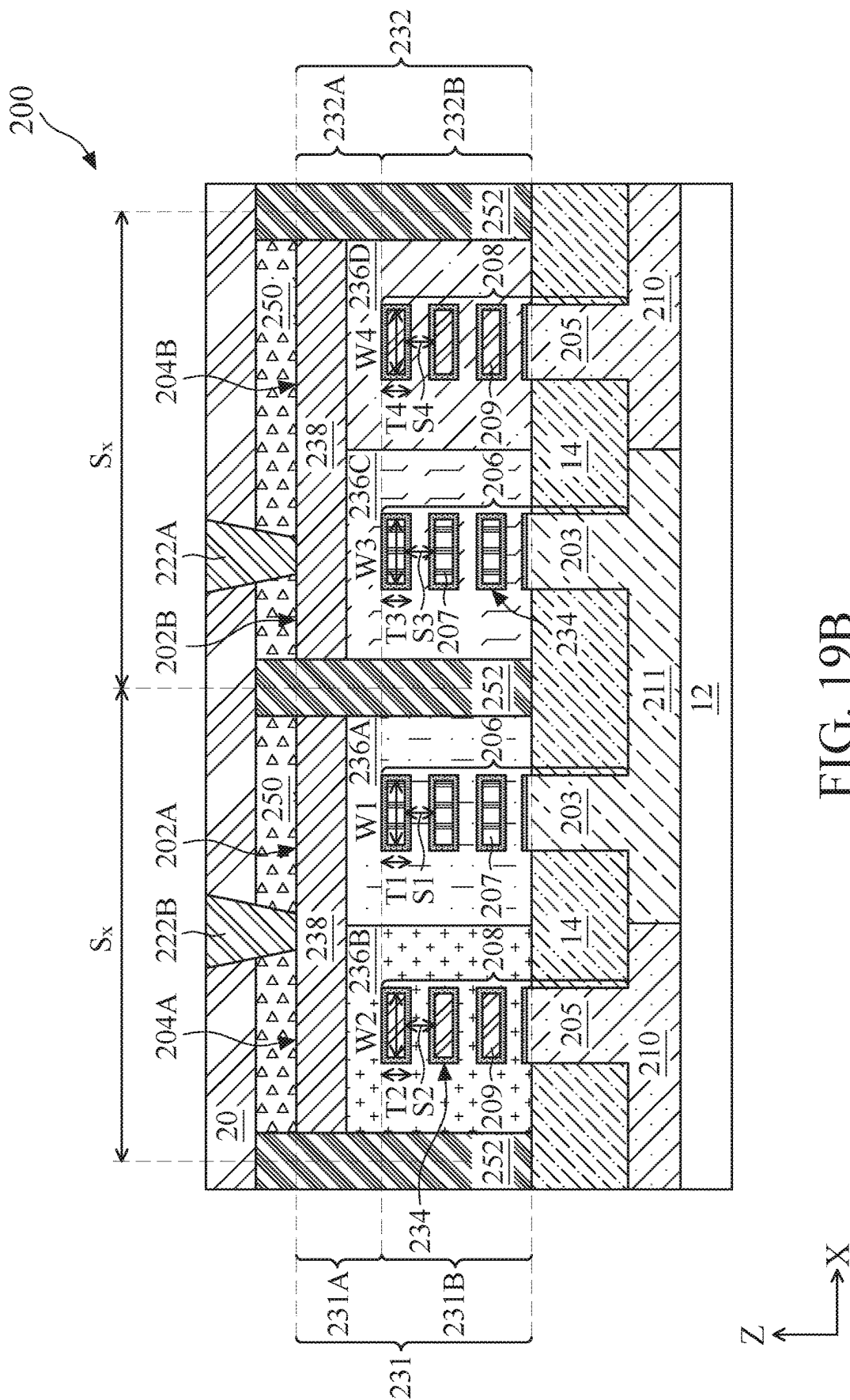


FIG. 19B

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METAL GATE STACKS AND METHODS OF FABRICATING THE SAME IN MULTI-GATE FIELD-EFFECT TRANSISTORS

PRIORITY DATA

This is a continuation application of U.S. application Ser. No. 17/377,578, filed Jul. 16, 2021, which is herein incorporated by reference in its entirety.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

As integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate transistors, such as nanostructure (NS) transistors, have been incorporated into memory devices (including, for example, static random-access memory, or SRAM, cells) and core devices (including, for example, standard logic, or STD, cells) to reduce chip footprint while maintaining reasonable processing margins. However, designing IC chips that include NS transistors for multiple applications involves complex and often-times costly processes. Accordingly, although existing technologies for fabricating NS transistors have been generally adequate for their intended purposes, they have not been entirely satisfactory in all aspects.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A is a diagrammatic plan view of an IC structure, in portion or entirety, according to various aspects of the present disclosure.

FIG. 1B is a diagrammatic plan view of a portion of the IC structure of FIG. 1A according to various aspects of the present disclosure.

FIG. 2 is a diagrammatic plan view of an IC device, in portion or entirety, according to various aspects of the present disclosure.

FIG. 3 is a cross-sectional view of the IC device of FIG. 2 taken along line AA' according to various aspects of the present disclosure.

FIG. 4 is a cross-sectional view of the IC device of FIG. 2 taken along line BB' according to various aspects of the present disclosure.

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FIG. 5 is a cross-sectional view of the IC device of FIG. 2 taken along line CC' according to various aspects of the present disclosure.

FIGS. 6A and 6B together illustrate a flow chart of a method for fabricating an IC device according to various aspects of the present disclosure.

FIG. 7 is a three-dimensional perspective view of an IC device according to various aspects of the present disclosure.

FIGS. 8A, 9, 10, 11, 12, 13A, 14A, 15A, 16A, 17A, 18A, and 19A are cross-sectional views of the IC device of FIG. 7 taken along line DD' according to various aspects of the present disclosure.

FIGS. 8B, 13B, 14B, 15B, 16B, 17B, 18B, and 19B are cross-sectional views of the IC device of FIG. 7 taken along line EE' according to various aspects of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, "lower," "upper," "horizontal," "vertical," "above," "over," "below," "beneath," "up," "down," "top," "bottom," etc. as well as derivatives thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features.

Furthermore, when a number or a range of numbers is described with "about," "approximate," and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within +/-10% of the number described or other values as understood by person skilled in the art. For example, the term "about 5 nm" encompasses the dimension range from 4.5 nm to 5.5 nm. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as three-dimensional nanostructure (NS) FETs (alternatively referred to as gate-all-around, or GAA, FETs), in memory and/or standard logic cells of an integrated circuit (IC) structure. Generally, an NS FET includes a plurality of vertically stacked sheets (e.g., nanosheets), wires (e.g., nanowires), or rods (e.g., nanorods) in a channel region of the FET, thereby allowing better gate control, lowered leakage current, and improved scaling capability for various IC applications. The present disclosure includes multiple embodiments. Different embodiments may have different advantages, and no particular advantage is necessarily required of any embodiment.

As length scales in semiconductor devices continue to decrease, reduced separation distance between vertically

stacked nanostructures (e.g., nanosheets, nanorods, nanowires) in an NS FET is desired for purposes of reducing resistance in S/D features as well as capacitance between the gate (e.g., a metal gate stack) and the S/D features of a device, among others. However, in some instances, tightened separation distance between vertically stacked nanostructures may limit the formation and the performance of a metal gate stack configured to wrap around each nanostructure. In one such example, tightened separation distance may lead to a lack of flexibility in tuning the threshold voltage (V_t) of the metal gate stack due to a reduced number of WFM layers that can be formed. In another example, tightened separation distance may lead to a potential increase in the resistance of the metal gate stack due to reduced thickness of the low-resistance metal fill (or bulk conductive) layer. Thus, for at least these reasons, improvements in the fabrication of metal gate stacks in NS FETs are desired.

Referring to FIGS. 1A and 1B, the present disclosure provides an IC structure (e.g., an IC chip) **100** formed over a semiconductor substrate and includes at least one IC device **200** (hereafter referred to as device **200**). The device **200** is provided for illustration purposes and does not necessarily limit the embodiments of the present disclosure to any number of transistors, any number of regions, or any configuration of structures or regions. Furthermore, the device **200**, or a portion thereof, may include memory devices (e.g., static random-access memory (SRAM), dynamic random-access memory (DRAM)), standard logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type FETs (PFETs), n-type FETs (NFETs), multi-gate FETs such as FinFETs and NS FETs, metal-oxide semiconductor field effect transistors (MOSFETs), complementary metal-oxide semiconductor FET (CMOSFET), bipolar transistors, high voltage transistors, high frequency transistors, other memory cells, and combinations thereof. Additional features can be added to the device **200** and/or the IC structure **100**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the IC structure **100**.

Referring to FIG. 1B, the device **200** includes at least two cells **201** disposed along the X-axis, where each cell **201** is configured to include a p-type three-dimensional fin-like active region **206** (hereafter referred to as p-type fin **206**) disposed in a p-type doped region **211** (hereafter referred to as p-well **211**) and a three-dimensional fin-like active region **208** (hereafter referred to as n-type fin **208**) disposed in an n-type doped region **210** (hereafter referred to as n-well **210**), where the n-well **210** is disposed adjacent to the p-well **211** within each cell **201**. In the present embodiments, p-wells **211** of the two adjacent cells **201** are disposed between the n-wells **210**. The p-type fins **206** and the n-type fins **208** are oriented lengthwise along the Y-axis and spaced from each other along the X-axis.

Each cell **201** may be defined by the long pitch S_x along the X-axis and the short pitch S_y along the Y-axis. In the present embodiments, the two cells **201** are depicted to be substantially similar in dimension, i.e., having substantially the same S_x and S_y . However, because different cells **201** may be directed to different applications, the cells **201** in the device **200** may differ in dimension and layout design.

Each cell **201** further includes a gate stack **231** and a gate stack **232** oriented substantially perpendicular to the p-type fins **206** and the n-type fins **208** (i.e., disposed along the Y-axis). In the present embodiments, adjacent gate stacks **231** and **232** along the X-axis are separated by gate isolation

features (or alternatively referred to as gate end isolation features) **252**. As will be discussed in detail below, an upper portion of each of the gate stacks **231** and **232** is disposed over a top surface of the p-type fins **206** and the n-type fins **208**, and a lower portion of each of the gate stacks **231** and **232** wrap around channel regions of the p-type fins **206** and the n-type fins **208**. In the depicted embodiments, the gate stacks **231** and **232** each include top gate spacers **216A** disposed on sidewalls of the upper portion (as depicted in FIG. 2) and inner gate spacers **216B** disposed on sidewalls of the lower portion (as depicted in FIG. 5). In some embodiments, the gate stacks **231** and **232** have substantially the same composition. In some embodiments, the gate stacks **231** and **232** differ in the composition of the gate electrode as discussed in detail below.

Various embodiments of portions of the device **200** are discussed in detail below. FIG. 2 shows a schematic planar top view of an embodiment of the device **200** (or a portion thereof), FIG. 3 is a schematic cross-sectional view of FIG. 2 taken along line AA', FIG. 4 is a schematic cross-sectional view of FIG. 2 taken along line BB', and FIG. 5 is a schematic cross-sectional view of FIG. 2 taken along line CC'. It is noted that FIGS. 2-5 have each been simplified for the sake of clarity to better illustrate embodiments of the present disclosure. As such, additional features can be added to the device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the cells **201**.

Referring now to FIGS. 2 and 3 collectively, each p-type fin **206** includes a first set of vertically stacked semiconductor layers **207** that engage with the gate stack **231** or **232** to form an n-type NS FET (hereafter referred to as a NS NFET), and each n-type fin **208** includes a second set of vertically stacked semiconductor layers **209** that engage with the gate stack **231** or **232** to form a p-type NS FET (hereafter referred to as a NS PFET). In the present embodiments, the semiconductor layers **207** and **209** are generally oriented lengthwise along the Y-axis and stacked vertically along the Z-axis. Furthermore, each stack of the semiconductor layers **207** is interposed between n-type source/drain (S/D) features **214N**, and each stack of the semiconductor layers **209** is interposed between p-type S/D features **214P** (as depicted in FIG. 4).

In the present embodiments, the semiconductor layers **207** engage with (or are wrapped around by) the gate stack **231** to form channel regions of an NS NFET **202A**, and the semiconductor layers **209** engage with (or are wrapped around by) the gate stack **231** to form channel regions of an NS PFET **204A**. Furthermore, the semiconductor layers **207** engage with the gate stack **232** to form channel regions of an NS NFET **202B**, and the semiconductor layers **209** engage with the gate stack **232** to form channel regions of an NS PFET **204B**. In other words, the semiconductor layers **207** are configured (and hereafter referred to) as channel layers **207** for the NS NFETs **202A** and **202B**, and the semiconductor layers **209** are configured (and hereafter referred to) as channel layers **209** for the NS PFETs **204A** and **204B**. In the present embodiments, the NS NFET **202A** and the NS PFET **204A** form a first NS CMOSFET and the NS NFET **202B** and the NS PFET **204B** form a second NS CMOSFET. In some embodiments, though not depicted herein, two n-type fins **208** (or two p-type fins **206**) may be disposed adjacent to each other and engage with the gate stack **231** to form two NS PFETs (or two NS NFETs). In the depicted embodiments, referring to FIGS. 2-4 for example, the NS NFET **202A** and the NS PFET **204A** share the common gate stack **231**, and the NS NFET **202B** and the NS PFET **204B** share

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the common gate stack **232**. As will be discussed in detail below, portions of the same gate stack configured to provide different NS FETs have different compositions according to some embodiments of the present disclosure.

Referring to FIG. 2, the device **200** further includes a plurality of S/D contacts **220A**, **220B**, **220C**, **220D**, and **220E** disposed on one or more S/D features and oriented lengthwise along the X-axis. In the present embodiments, the S/D contacts **220A** and **220B** are configured to contact p-type S/D features **214P** disposed over (or in) the n-type fins **208**, and the S/D contacts **220C**, **220D**, and **220E** are configured to contact n-type S/D features **214N** disposed over (or in) the p-type fins **206**. In some embodiments, different S/D contacts **220A-220E** may vary in length along the X-axis. For example, the S/D contact **220B** is longer than the S/D contact **220A**, the S/D contact **220D** is longer than the S/D contact **220C**, and the S/D contact **220E** is longer than the S/D contact **220D**. In the depicted embodiments, the S/D contacts **220E** may be electrically coupled to the n-type S/D features **214N** of two adjacent NS NFETs (**202A** and **202B**) and further connected to a vertical interconnect structure (e.g., via **226B**). The S/D contacts **220A-220E** may alternatively be referred to as device-level contacts to be differentiated from other contact features (e.g., vias and conductive lines) subsequently formed as portions of a multi-layer interconnect (MLI) structure over the device **200**.

Still referring to FIG. 2, the device **200** may further include a plurality of vertical interconnect structures (or vias) configured to electrically connect various NS FETs with a subsequently formed metal layer (not depicted) and/or to electrically connect two metal layers together. In the depicted embodiments, the device **200** includes vias **226A** configured to electrically connect one or more of the device-level S/D contacts **220A-220D** with a subsequently formed metal layer and a via **226B** configured to electrically connect two adjacent S/D contacts **220D** together. In the present embodiments, the device **200** further includes a plurality of gate contacts **222A** and **222B** disposed over portions of the gate stacks **231** or **232**. As depicted herein, the gate contacts **222A** and **222B** differ in their location with respect to the channel regions of the NS FETs of the device **200**. For example, the gate contacts **222A** are disposed on portions of the gate stacks **231** or **232** that are directly above the channel regions (i.e., the stacks of channel layers **207** or **209**), while the gate contacts **222B** are disposed on portions of the gate stacks **231** or **232** that are directly above isolation structures **14** disposed over a semiconductor substrate (or a wafer; hereafter referred to as substrate) **12** on which the device **200** is formed.

Referring to FIG. 3, components of the device **200** are disposed over the substrate **12**, which includes n-wells **210** and p-wells **211** over which the n-type fins **208** and p-type fins **206** are formed, respectively. The device **200** further includes the isolation structures **14** disposed over the substrate **12** to electrically separate various active regions formed over the substrate **12**. In the present embodiments, the isolation structures **14** include shallow trench isolation (STI) features. In the depicted embodiments, each stack of the channel layers **207** is disposed over a base fin **203** and each stack of the channel layers **209** is disposed over a base fin **205**, where the base fins **203** and **205** protrude from the substrate **12** and are separated by the isolation structures **14**.

Each of the channel layers **207** and **209** may include Si, Ge, SiC, SiGe, GeSn, SiGeSn, SiGeCSn, other suitable semiconductor materials, or combinations thereof. In the present embodiments, each of the semiconductor layers **207**

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and **209** includes elemental Si in the form of a nanosheet, a nanowire (e.g., a nanowire having a hexagonal cross-section), a nanorod (e.g., a nanorod having a square or circular cross-section), or other suitable configurations. In some embodiments, the p-type fin **206** and the n-type fin **208** each include two to ten channel layers **207** and **209**, respectively. In the present embodiments, the p-type fin **206** and the n-type fin **208** each include no more than four channel layers **207** and **209**, respectively. For example, the p-type fin **206** and the n-type fin **208** may each include three channel layers **207** and three channel layers **209**, respectively. Of course, the present disclosure is not limited to such configurations and the number of semiconductor layers may be tuned according to design requirements for the device **200**.

Still referring to FIG. 3, each stack of the channel layers **207** and **209** may be defined by a width of the stack measured along the lengthwise direction of the gate stacks **231** and **232**, a layer thickness measured along the stacking direction of the channel layers, and a separation distance between adjacent layers. For example, each stack of the channel layers **207** engaged with the gate stack **231** has a width of W_1 , a layer thickness of T_1 , and a layer separation distance of S_1 , and each stack of the channel layers **209** engaged with the gate stack **231** has a width of W_2 , a layer thickness of T_2 , and a layer separation distance of S_2 . Similarly, each stack of the channel layers **207** engaged with the gate stack **232** has a width of W_3 , a layer thickness of T_3 , and a layer separation distance of S_3 , and each stack of the channel layers **209** engaged with the gate stack **232** has a width of W_4 , a layer thickness of T_4 , and a layer separation distance of S_4 . In the present embodiments, T_1 , T_2 , T_3 , and T_4 are substantially the same in magnitude, and S_1 , S_2 , S_3 , and S_4 are substantially the same in magnitude. In some examples, the sheet thickness T_1 , T_2 , T_3 , and T_4 may each be about 4 nm to about 8 nm and the sheet separation distance S_1 , S_2 , S_3 , and S_4 may each be about 6 nm to about 15 nm. In some embodiments, W_1 , W_2 , W_3 , and W_4 are substantially the same in magnitude. In some embodiments, W_2 is greater than W_1 and W_4 is greater than W_3 . For example, a ratio of W_2 to W_1 and a ratio of W_4 to W_3 may each be about 1.05 to about 2. It is noted that “substantially the same” as used in the present disclosure refers to a difference within about $\pm 5\%$ between two values. Of course, other dimensions of the stack width, the layer thickness, and the layer separation distance may also be applicable to embodiments of the present disclosure.

In the present embodiments, as will be discussed in detail below, the gate stack **231** includes a top portion **231A** disposed over a bottom portion **231B** and the gate stack **232** includes a top portion **232A** disposed over a bottom portion **232B**, where the top and the bottom portions of each gate stack differ in composition. Furthermore, portions of the same gate stack configured to form different FET devices, e.g., the NS NFETs and the NS PFETs, also differ in composition required by design criteria.

Referring to the gate stack **231** depicted in FIG. 3, the bottom portion **231B** includes a gate dielectric layer **234** wrapping around each channel layer **207** and **209** of the NS NFET **202A** and the NS PFET **204A**, respectively. In the present embodiments, an additional portion of the gate dielectric layer **234** is disposed over the base fins **203** and **205**, respectively. The gate dielectric layer **234** may include any suitable material, such as silicon oxide, silicon oxynitride, a high-k dielectric material (i.e., having a dielectric constant greater than that of silicon oxide, which is about 3.9) such as hafnium oxide, tantalum oxide, aluminum oxide, zirconium oxide, lanthanum oxide, titanium oxide,

yttrium oxide, strontium titanate, other suitable dielectric materials, or combinations thereof. In some embodiments, the gate dielectric layer **234** has a dielectric constant k of greater than about 9. In some examples, the gate dielectric layer **234** may be about 0.5 nm to about 3 nm in thickness; though the present embodiments are not limited to such dimension. In some embodiments, the bottom portion **231B** includes an interfacial layer (IL; not depicted) disposed between the p-type fin **206** and the n-type fin **208** (including their respective channel layers and base fin) and the gate dielectric layer **234**, where the IL includes an oxide, such as silicon oxide.

The bottom portion **231B** further includes a work-function metal (WFM) layer **236A** and a **236B** disposed over the gate dielectric layer **234** and configured to provide the NS NFET **202A** and the NS PFET **204A**, respectively. In the present embodiments, each of the WFM layers **236A** and **236B** completely fills the space disposed between two vertically stacked channel layers **207** and **209**, respectively. In other words, each of the WFM layers **236A** and **236B** defines a sidewall of the bottom portion **231B** of the gate stack **231**. Accordingly, in the depicted embodiments, the WFM layers **236A** and **236B** extend laterally to directly contact the gate isolation features **252**, which separate the gate stack **231** from an adjacent gate stack, such as the gate stack **232**. In other words, each gate isolation feature **252** defines an outer sidewall of each of the WFM layers **236A** and **236B**. Notably, by configuring the space between the channel layer **207** and **209** to be filled completely with the WFM layers **236A** and **236B**, respectively, the present embodiments provide greater design flexibility in tuning the V_t of the gate stack **231**, improving performance of the NS FETs as a result.

Each of the WFM layers **236A** and **236B** may include one or more WFM, such as TiN, TaN, TiAl, TaAl, TaAlC, TiAlN, TiAlC, TaC, TaCN, TaSiN, WN, WNC, other suitable WFMs, or combinations thereof. In the present embodiments, each of the WFM layers **236A** and **236B** is substantially free of W, Cu, Ru, Co, or combinations thereof. In some embodiments, specific compositions of the WFM layers **236A** and **236B** are selected based on an overall work function desired for their respective FETs. In some embodiments, the WFM layer **236A** and **236B** differ in composition, such that the work function of the portion of the gate stack **231** configured to form the NS NFET **202A** is different from the work function of the portion of the gate stack **231** configured to form the NS PFET **204A**. In some embodiments, the difference in composition between the WFM layers **236A** and **236B** allows the V_t of the resulting NS CMOSFET to be tuned according to a given design requirement. In some embodiments, the gate stack **231** includes additional material layer, such as a barrier layer, disposed between the gate dielectric layer **234** and the WFM layer **236A** and/or **236B**.

As discussed above, the top portion **231A** of the gate stack **231** is disposed over the bottom portion **231B**. In the present embodiments, the top portion **231A** includes a metal fill layer (also referred to as a bulk conductive layer) **238** disposed over portions of the WFM layers **236A** and **236B** that extend beyond a top surface of the p-type fin **206** and the n-type fin **208**. In other words, a bottom surface of the metal fill layer **238** defines the top surface of both the WFM layers **236A** and **236B**. In the present embodiments, sidewalls of the metal fill layer **238** are continuous with the outer sidewalls of the WFM layers **236A** and **236B** along the Z-axis, such that the sidewalls of the metal fill layer **238** are in direct contact with the gate isolation features **252**, which

isolates the gate stack **231** from adjacent gate stacks. Taken as a whole, the gate stack **231** includes at least a gate dielectric layer **234** wrapping around each channel layer **207** and **209**, the WFM layers **236A** and **236B** disposed over the gate dielectric layer **234** and filling the space between two vertically stacked channel layers **207** and **209**, respectively, and the metal fill layer **238** disposed over the top surfaces of the WFM layers **236A** and **236B**. Accordingly, the portion of the gate stack **231** providing the NS NFET **202A** and the portion of the gate stack **231** providing the NS PFET **204A** share the same metal fill layer **238** but differ in the composition of the WFM layers. Notably, the present embodiments provide that the bottom portion **231B** of the gate stack **231** is free of the metal fill layer **238**, which is included in the top portion **231A** of the gate stack **231**. In other words, the space between the channel layers **207** and **209** is free of the metal fill layer **238**.

For the top portion **231A**, the metal fill layer **238** may be defined by a thickness $H1$ measured along the stacking direction (i.e., the Z-axis) of the channel layers **207** and **209**, and the WFM layers **236A** and **236B** may each be defined by a thickness $H2$. In some embodiments, $H1$ is at least the same as $H2$. It is noted that the present embodiments are not limited to such dimension so long as both $H1$ and $H2$ are greater than zero. In some instances, $H1$ and $H2$ may each be about 2 nm to about 20 nm. In some instances, $H1$ and $H2$ may each be about 4 nm to about 12 nm. As discussed in detail below, the thickness $H1$ of the metal fill layer **238** may be increased without needing to enlarge the layer separation distance (e.g., $S1$ and $S2$), thereby reducing the overall resistance of the metal gate stack **231**.

In the present embodiments, the metal fill layer **238** includes W, Cu, Ru, Co, or combinations thereof and is substantially free of any of the WFM material discussed above with respect to the WFM layers **236A** and **236B**. In an example embodiment, the metal fill layer **238** includes W. In some embodiments, the composition of the metal fill layer **238** is selected such that the resistance of the metal fill layer **238** is less than the resistance of each of the WFM layers **236A** and **236B**.

Still referring to FIG. 3, the device **200** further includes an etch-stop layer (ESL) **250** disposed over the gate stack **231**, i.e., over the metal fill layer **238**, and an interlayer dielectric (ILD) layer **20** disposed over the ESL **250**. The ESL **250** may include any suitable dielectric material, such as silicon nitride, silicon carbide, oxygen-doped silicon nitride (SiON), carbon-doped silicon nitride (SiCN), aluminum nitride, other suitable materials, or combinations thereof. The ILD layer **20** may include a low- k dielectric material, silicon oxide, doped silicate glass, other suitable materials, or combinations thereof. In the present embodiment, the ESL **250** and the ILD layer **20** differ in composition to ensure adequate etching selectivity therebetween during subsequent fabrication process.

As discussed above, the device **200** further includes the gate contacts **222A** and **222B** configured to electrically couple the gate stacks **231** and **232** with subsequently formed interconnect features. In the depicted embodiments, the gate contact **222A** is disposed over the channel regions of any one or more of the NF FETs of the device **200**, while the gate contact **222B** is vertically offset from the channel regions of any one or more of the NF FETs and is instead disposed over the isolation structures **14**. Notably, referring to FIG. 3, the present embodiments provide that a width W of the metal fill layer **238** spans across the entire width of the gate stack **231** between the gate isolation features **252** along the X-axis, which allows greater freedom in determining

locations of the gate contacts **222A** and **222B** that are suitable for improving routing efficiency of the device **200**.

Still referring to FIG. 3, the present embodiments provide that the gate stacks **231** and **232** have substantially the same structural arrangement. For example, the gate stack **232** includes the top portion **232A** disposed over the bottom portion **232B**, where the bottom portion **232B** includes the gate dielectric layer **234** wrapping around each channel layer **207** and **209**, WFM layers **236C** and **236D** disposed over the gate dielectric layer **234** and filling the space between two vertically stacked channel layers **207** and **209**, and where the top portion **232A** includes the metal fill layer **238** disposed over the top surfaces of the WFM layers **236C** and **236D**. In other words, the top portion **232A** includes the metal fill layer **238** but the bottom portion **232B** is free of the same. In some examples, the composition of the WFM layer **236C** may be substantially the same as or different from the WFM layer **236A**, which is discussed above with respect to the gate stack **231**. Independently, the composition of the WFM layer **236D** may be substantially the same as or different from the WFM layer **236B**, which is also discussed above with respect to the gate stack **231**. In the present embodiments, the composition of the WFM layers **236C** and **236D** are selected based on an overall work function desired for their respective FETs. For example, the WFM layer **236C** and **236D** may differ in composition, such that the work function of the portion of the gate stack **232** configured to form the NS NFET **202B** is different from the work function of the portion of the gate stack **232** configured to form the NS PFET **204B**, thereby allowing the V_t of the resulting NS CMOSFET to be tuned according to a given design requirement. Furthermore, for the top portion **232A**, the metal fill layer **238** is defined by a thickness $H3$ in the gate stack **232** and the portions of the WFM layers **236C** and **236D** disposed above the channel layers **207** and **209** are defined by a thickness $H4$, where $H3$ and $H4$ are similar to $H1$ and $H2$, respectively, i.e., $H3 \approx H4$, as discussed in detail above.

Generally, reducing the layer separation distance (e.g., $S1$, $S2$, $S3$, and $S4$) in an NS FET lowers the overall height of the channel region (or the height of the gate stack), such that both the parasitic capacitance between the gate stack and the S/D features as well as the resistance of the S/D features may be reduced for improved device performance. However, reduction in layer separation distance introduces challenges in the fabrication of the metal gate stack, such as limitation in the number of WFM layers and the thickness of the low-resistance metal fill layer that can be accommodated between the channel layers. The present embodiments provide methods of forming a metal gate stack with improved flexibility in V_t tuning and ability to accommodate fabrication of low-resistance metal fill layer.

Now turning to FIG. 4, which is a cross-sectional view of FIG. 2 taken along line BB', i.e., through one of the n-type fins **208** along the Y-axis. In the depicted embodiments, each gate stack **231** (or **232**) is disposed between two p-type S/D features **214P** along the Y-axis, where the top portion **231A** (or **232A**) of the gate stack **231** (or **232**) is disposed over the bottom portion **231B** (or **232B**), i.e., over the topmost channel layer **209**. Stated differently, the bottom portion **231B** of the gate stack **231** is interleaved with the channel layers **209**. In the present embodiments, the bottom portion **231B** includes the gate dielectric layer **234** disposed over the channel layers **209** (and in contact with the inner gate spacers **216B**) and the WFM layer **236B** over the gate dielectric layer **234** and filling the space between the channel layers **209**. Notably, the bottom portion **231B** is free of the metal fill layer **238** as discussed above with respect to FIG.

3. The top portion **231A** includes the gate dielectric layer **234**, the WFM layer **236B** disposed over the gate dielectric layer **234**, and the metal fill layer **238** disposed over the top surface of the WFM layer **236B**. In the present embodiments, the gate dielectric layer **234** is disposed along and define the sidewalls of both the WFM layer **236B** and the metal fill layer **238** in the top portion **231A**. As discussed above, the device **200** further includes the ESL **250** disposed over the top surface of the gate stack **231** to accommodate subsequent fabrication of components such as the S/D contacts **220A** and **220B**. In some embodiments, a bottom portion of the p-type S/D features **214P** extends to below the bottommost channel layer **209** by a distance **215** of about 3 nm to about 40 nm.

In the present embodiments, still referring to FIG. 4, the device **200** further includes the top gate spacers **216A** disposed along the sidewalls of the top portion **231A**, and the inner gate spacers **216B** disposed on sidewalls on the gate dielectric layer **234** to separate the bottom portion **231B** of the gate stack **231** from the adjacent p-type S/D features **214P**. Each of the top gate spacers **216A** and the inner gate spacers **216B** may be a single-layer structure or a multi-layer structure and may include silicon oxide, silicon nitride, silicon carbide, oxygen-doped silicon nitride (SiON), carbon-doped silicon nitride (SiCN), oxygen-and-carbon-doped silicon nitride (SiOCN), oxygen-doped silicon carbide (SiOC), a low-k dielectric material, tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), other suitable materials, or combinations thereof. In some embodiments, the top gate spacers **216A** and the inner gate spacers **216B** have different compositions. In some embodiments, the top gate spacers **216A** and/or the inner gate spacers **216B** includes an air gap.

Still referring to FIG. 4, the device **200** may further include the S/D contacts **220A** and **220B** disposed over and electrically contact the p-type S/D features **214P**. The device **200** may include a silicide layer **223** disposed between each of the S/D contacts **220A** (and **220B**) and the p-type S/D feature **214P**. The silicide layer **223** may include nickel silicide, titanium silicide, cobalt silicide, other suitable silicides, or combinations thereof. The device may further include the via **226A** disposed in the ILD layer **20** and over one or more of the S/D contacts **220A** and **220B**, thereby interconnecting the S/D contact with subsequently formed MLI structure over the device **200**.

Now referring to FIG. 5, which is a cross-sectional view of FIG. 2 taken along line CC', i.e., through S/D regions of the NS NFETs (**202A** and **202B**) and the NS PFETs (**204A** and **204B**) along the X-axis. In the present embodiments, the NS NFETs **202A** and **202B** each include the n-type S/D feature **214N** grown over the base fin **203** in the S/D region of the p-type fin **206**, and the NS PFETs **204A** and **204B** each include the p-type S/D feature **214P** grown over the base fin **205** in the S/D region of the n-type fin **208**. The n-type S/D features **214N** and the p-type S/D features **214P** each include one or more epitaxially grown semiconductor layer doped with a suitable dopant. The n-type S/D features **214N** may include one or more epitaxial layers of silicon (epi Si) or silicon carbon (epi SiC) doped with an n-type dopant such as arsenic, phosphorus, other n-type dopants, or combinations thereof. The p-type S/D features **214P** may include one or more epitaxial layers of silicon germanium (epi SiGe) doped with a p-type dopant such as boron, germanium, indium, other p-type dopants, or combinations thereof. In some embodiments, the n-type S/D features **214N**

and the p-type S/D features **214P** are grown from and wrap around at least a top portion of base fins **203** and **205**, respectively. In some examples, adjacently disposed S/D features may be merged, such that their widths along the X-axis span over more than one fins.

FIGS. **6A** and **6B** illustrates a method **400** and a method **500** for forming the device **200**, or portions thereof, in accordance with some embodiments of the present disclosure. The methods **400** and **500** are discussed in reference to FIGS. **7-18B**, where FIG. **7** is a three-dimensional perspective view of the device **200**, FIGS. **8A, 9-12, 13A, 14A, 15A, 16A, 17A, 18A, and 19A** are cross-sectional views of FIG. **7** taken along line DD', and FIGS. **8B, 13B, 14B, 15B, 16B, 17B, 18B, and 19B** are cross-sectional views of FIG. **7** taken along line EE', all at intermediate steps of the methods **400** and/or **500**. It is noted that although the methods **400** and **500** are discussed in reference to the cross-sectional views of one of the n-type fins **208** as depicted in FIGS. **8A, 9-12, 13A, 14A, 15A, 16A, 17A, 18A, and 19A**, the p-type fin **206** undergoes the substantially the same operations (which may be implemented together or separately) in accordance with the embodiments provided herein. The methods **400** and **500** are merely examples and are not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be performed before, during, and after the method **400** or the method **500**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the methods.

Operations **402-410** of the method **400** are discussed in reference to FIGS. **7, 8A, and 8B** collectively. At operation **402**, the method **400** provides a workpiece, such as the IC structure **100**, that includes the substrate **12** and various doped regions (e.g., the n-wells **210** and p-wells **211**) formed in or over the substrate **12**. In the present embodiments, the substrate **12** includes silicon. Alternatively or additionally, the substrate **12** includes another elementary semiconductor, such as germanium; a compound semiconductor, such as silicon carbide, silicon phosphide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor, such as SiGe, SiPC, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. In some embodiments, the substrate **12** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, other suitable methods, or combinations thereof.

Each n-well **210** may be doped with an n-type dopant, such as phosphorus, arsenic, other n-type dopants, or combinations thereof. Each p-well **211** may be doped with a p-type dopant, such as boron, indium, other p-type dopants, or combinations thereof. In some embodiments, the substrate **12** includes doped regions formed with a combination of p-type dopants and n-type dopants. The various doped regions can be formed directly on and/or in the substrate **12** providing, for example, a p-well structure, an n-well structure, a dual-well structure, a raised structure, or combinations thereof. Each of the various doped regions may be formed by performing an ion implantation process, a diffusion process, other suitable doping processes, or combinations thereof.

At operation **404**, the method **400** forms a multi-layer stack of semiconductor materials (hereafter referred to as the "multi-layer stack" for short) over the substrate **12**. In the present embodiments, the p-type fins **206** and the n-type fins

208 are formed from the multi-layer stack at subsequent operations of the method **400**. In the present embodiments, the multi-layer stack includes alternating layers of a first semiconductor material (e.g., epitaxially grown Si-containing layers **207** and **209** that are substantially free of Ge) and a second semiconductor material (e.g., epitaxially grown SiGe-containing layers **241** and **243**) grown in a series of epitaxy processes. The epitaxy process may include CVD deposition techniques (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), low-pressure CVD (LP-CVD), and/or plasma-enhanced CVD (PE-CVD)), molecular beam epitaxy, other suitable selective epitaxial growth (SEG) processes, or combinations thereof. The epitaxy process may use gaseous and/or liquid precursors, which interact with the composition of the underlying material layers. In some examples, the layers of the multi-stack may be provided in the form of nanosheets, nanowires, or nanorods. Subsequent processing may remove the second semiconductor layers (e.g., the SiGe-containing layers **241** and **243**), leaving behind the first semiconductor layers (e.g., the Si-containing layers **207** and **209**) separated by openings. Such a process, which will be discussed in detail below, may be referred to as the "wire release process" or "sheet formation process," depending upon the configuration of the layers in the multi-layer stack. In the present embodiments, the remaining stack of Si-containing layers **207** and **209** become the channel layers configured to form a NS NFET (**202A** or **202B**) and a NS PFET (**204A** or **204B**), respectively, in the device **200**, and the removed SiGe-containing layers **241** and **243** are hereafter referred to as the non-channel layers.

At operation **406**, the method **400** forms the p-type fins (or p-type fin active regions) **206** and the n-type fins (or n-type fin active regions) **208** from the multi-layer stack. Accordingly, the semiconductor fins of the present embodiments include alternating layers of Si (**207** or **209**) and SiGe (**241** or **243**) as discussed above. In some embodiments, the device **200** additionally includes semiconductor fins (not depicted herein) having a single semiconductor material rather than alternating layers of different semiconductor materials. The p-type fins **206** and the n-type fins **208** may be fabricated by directly patterning and subsequently etching the multi-layer stack having alternating layers of epitaxially grown Si- and SiGe-containing layers. The fabrication process may include forming a masking element including a photoresist layer, lithographically patterning the masking element, and subsequently etching the multi-layer stack (and portions of the substrate **12**) using the patterned masking element as an etch mask. The etching process may include dry etching, wet etching, reactive ion etching (RIE), other suitable processes, or combinations thereof. The resulting p-type fins (or fin active regions) **206** and the n-type fins **208** may be doped with various dopants consistent with desired design requirements.

Additionally or alternatively, other embodiments of methods for forming p-type fins **206** and the n-type fins **208** may also be suitable. For example, the multi-layer stack (and the substrate **12**) may be patterned using double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial

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layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the multi-layer stack to form the p-type fins **206** and the n-type fins **208**.

At operation **408**, the method **400** forms the isolation structures **14** to insulate various components formed over the substrate **12**. The isolation structures **14** may include STI, field oxide, LOCAl oxidation of silicon (LOCOS), other suitable features comprising silicon oxide, silicon nitride, silicon oxynitride, other suitable dielectric materials, or combinations thereof. The isolation structures **14** may be formed by any suitable method. In some embodiments, the isolation structures **14** are formed by filling trenches formed between the semiconductor fins in the multi-layer stack with a dielectric material, followed by applying a chemical mechanical planarization (CMP) process and an etch-back process. In some embodiments, the isolation structures **14** are formed by depositing a dielectric material over sidewalls of the p-type fins (or fin active regions) **206** and the n-type fins **208** without completely filling the trenches between them. The isolation structures **14** may include a multi-layer structure, for example, having one or more thermal oxide liner layers.

At operation **410**, still referring to FIGS. 7-8B, the method **400** forms a dummy gate stack (or a placeholder gate) **246** over the p-type fins **206** and the n-type fins **208**. The dummy gate stack **246** may include, for example, an IL (including, for example, silicon oxide) formed over the p-type fins **206** and the n-type fins **208** and a dummy gate electrode layer (including, for example, polysilicon) formed over the IL. After forming other components (e.g., the n-type S/D features **214N** and the p-type S/D features **214P**) of the device **200**, portions of the dummy gate stack **246** (e.g., the dummy gate electrode layer) is removed to form a gate trench in which at least a gate dielectric layer (e.g., the gate dielectric layer **234**) and a metal gate electrode (e.g., including the WFM layers **236A-236D** and the metal fill layer **238**) are subsequently formed to complete the fabrication of the gate stacks **231** and **232**. Various material layers of the dummy gate stack **246** may be first deposited as a blanket layer over the semiconductor fins and subsequently patterned, followed by one or more etching process, to form the dummy gate stack **246** in a desired configuration in the device **200**. The various material layers of the dummy gate stack **246** may be formed by any suitable method, such as chemical oxidation, thermal oxidation, CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), electroplating, other suitable methods, or combinations thereof.

The method **400** proceeds to forming the top gate spacers **216A** on sidewalls of the dummy gate stack **246** at operation **410**. The top gate spacers **216A** may be a single-layer structure or a multi-layer structure and may include any suitable dielectric material discussed above with respect to FIG. 4. The top gate spacers **216A** may be formed by first depositing at least one spacer layer over the device **200** and subsequently performing an anisotropic etching process to the spacer layer, leaving behind the top gate spacers **216A** on the sidewalls of the dummy gate stack **246**.

At operation **412**, referring to FIGS. 9 and 10, the method **400** forms the inner gate spacers **216B** on portions of the multi-layer stack exposed in S/D recesses **260**. Referring to FIG. 9, the method **400** first removes portions of the n-type fin **208** (and the p-type fin **206**) to form the S/D recesses **260** and expose sidewalls of the multi-layer stack therein. Subsequently, the method **400** forms the inner gate spacers **216B** on the exposed sidewalls of the non-channel layers **243** (and the non-channel layers **241**). In some embodiments, forming the inner gate spacers **216B** includes selectively removing

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portions of the non-channel layers **243** without removing (or substantially removing) the channel layers **209** to form openings (not depicted), depositing a spacer layer in the openings, and performing one or more etch-back process to form the inner gate spacers **216B** in the openings (FIG. 10). The inner gate spacers **216B** may be a single-layer structure or a multi-layer structure and may include any suitable dielectric material discussed above with respect to FIG. 4.

At operation **414**, referring to FIG. 11, the method **400** forms the p-type S/D features **214P** in the S/D recesses **260**. The composition of the p-type S/D feature **214P** has been discussed above with respect to FIG. 5. In the present embodiments, one or more epitaxy growth processes similar to those discussed above with respect to forming the multi-stack layer are performed to grow the p-type S/D features **214P**. In some embodiments, the p-type S/D features **214P** includes an epitaxial material, such as epi SiGe, doped in-situ by adding a suitable dopant during the epitaxy growth process. In some embodiments, the epitaxial material is doped by an ion implantation process after performing a deposition process. In some embodiments, annealing processes are performed to activate dopants in the p-type S/D features **214P** and/or other S/D regions, such as HDD regions and/or LDD regions.

At operation **416**, referring to FIGS. 12, 13A, and 13B, the method **400** removes the dummy gate stack **246** to form a gate trench **262** between the top gate spacers **216A**. In the present embodiments, referring to FIG. 12, forming the gate trench **262** includes forming an ILD layer **18** over the p-type S/D features **214P**. The ILD layer **18** may have a composition similar to that of the ILD layer **20** discussed above with respect to FIG. 4. For example, the ILD layer **18** may include a low-k dielectric material, silicon oxide, doped silicate glass, other suitable materials, or combinations thereof, and may be formed by any suitable method, such as spin-on-glass or flowable CVD (FCVD). A top surface of the ILD layer **18** may be planarized using one or more CMP process. Thereafter, referring to FIGS. 13A and 13B, the method **400** proceeds to removing the dummy gate stack **246** from the device **200** by any suitable method, such as a dry etching process and/or a wet etching process, resulting in the gate trench **262** between the top gate spacers **216A**.

At operation **418**, still referring to FIGS. 13A and 13B, the method **400** removes the non-channel layers **243** (e.g., the SiGe-containing layers) from the multi-layer stack by one or more etching process, such as a dry etching process and/or a wet etching process, to leave behind the channel layers **209** (e.g., the Si-containing layers). In other words, after removing the non-channel layers **243**, openings **264** are inserted between or interleaved with the stack of channel layers **209**. In the present embodiments, the one or more etching process selectively remove the non-channel layers **243** without removing or substantially removing the channel layers **209**, respectively.

At operation **420**, referring to FIGS. 14A-17B collectively, the method **400** forms a gate stack **230**, which is subsequently separated or cut at operation **424** (see FIGS. 19A and 19B) into the gate stacks **231** and **232** as discussed above. In the present embodiments, referring to FIG. 6B, forming the gate stack **230** is implemented by the method **500**. Structures and compositions of various components of the gate stack **230** consistent with those of the gate stacks **231** and **232** share the same reference numerals as those provided above with respect to FIG. 4 and will thus be omitted for the discussion of the method **500** below for purposes of simplicity.

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Referring to FIGS. 14A and 14B, the method 500 at operation 502 first forms the gate dielectric layer 234 in the gate trench 262 and the openings 264, such that the gate dielectric layer 234 is deposited on surfaces of each channel layer 207 and 209 as well as the top surfaces of the base fins 203 and 205. The gate dielectric layer 234 may be formed by ALD, CVD, other suitable processes, or combinations thereof. In the present embodiments, the gate dielectric layer 234 is deposited conformally on each channel layer 207 and 209. In some embodiments, the method 500 first forms an IL (not depicted) over the channel layers 207 and 209 before depositing the gate dielectric layer 234.

Still referring to FIGS. 14A and 14B, the method at operation 504 forms the WFM layers 236A-236D over the gate dielectric layer 234. In the present embodiments, forming the WFM layers 236A-236D completely fills the gate trench 262 and the openings 264, such that the outer sidewalls of the WFM layers 236A and 236D define the sidewalls of the gate stack 230 when viewed along a lengthwise direction of the gate stack 230 (FIG. 14B). In some embodiments, the method 500 forms the WFM layers 236A-236D via a series of deposition and patterning processes, such that the composition, and the work function, of each of the WFM layers 236A-236D is suitable for tuning the V_t of their respective NS FETs. The deposition and patterning processes may be repeated if the WFM layers 236A-236D each include more than one WFMs. Of course, the present embodiments do not limit the order in which the WFM layers 236A-236D are formed. For example, the method 500 may form the WFM layer 236B first and proceed to patterning the WFM layer 236B and forming the WFM layer 236A subsequently. One or more CMP process may be implemented thereafter to planarize the top surface of the WFM layers 236A-236D. Each of the WFM layers 236A-236D may be deposited by ALD, CVD, PVD, other suitable deposition processes, or combinations thereof.

Now referring to FIGS. 15A and 15B, the method 500 at operation 506 recesses the WFM layers 236A-236D to form a trench 270. In the present embodiments, recessing the WFM layers 236A-236D does not completely remove portions of the WFM layers 236A-236D disposed over the channel layers 207 and 209. Defining the trench 270 to have a depth H1 (which also defines the thickness of the metal fill layer 238) and the remaining portions of the WFM layers 236A-236D to have the thickness H2 as discussed above, the present embodiments provide that H2 is greater than 0. In some examples, H2 may be less than H1; though the present embodiments are not limited thereto. The WFM layers 236A-236D may be recessed by an etching process 604, which may be a dry etching process and/or a wet etching process, using an etchant including a fluorine-based etchant (e.g., CF_4 , SF_6), a chlorine-based etchant (e.g., CCl_4), hydrogen peroxide, an acid (e.g., H_3PO_4 , HNO_3), other suitable etchants, or combinations thereof, where the type of etchants used for the etching process 604 may be selected based on the compositions of the WFM layer 236A-236D. In some embodiments, the depth H1 may be controlled by adjusting various parameters of the etching process 604 including, for example, etching duration.

Subsequently, referring to FIGS. 16A-17B, the method 500 at operation 508 forms the metal fill layer 238 in the trench 270, thereby filling the gate trench 262. Referring to FIGS. 16A and 16B, the metal fill layer 238 may be formed by a deposition process 606, which may include CVD, ALD, PVD, plating, other processes, or combinations thereof. In the present embodiments, the method 500 deposits W in the trench 270 to form the metal fill layer 238. In some embodi-

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ments, the deposition process 606 is an epitaxial growth process during which the metal fill layer 238 selectively grows from a metal-containing surface, e.g., the WFM layers 236A-236D, rather than from a dielectric-containing surface, e.g., the ILD layer 18 or the top gate spacers 216A. In other words, the deposition process 606 forms the metal fill layer 238 in a substantially bottom-up growth scheme in the trench 270. In some embodiments, such bottom-up growth scheme causes a top portion 620 of the metal fill layer 238 to protrude from a top surface of the ILD layer 18. Accordingly, the method 500 at operation 508 subsequently performs one or more CMP process along line 608 to remove the protruded top portion 620, thereby planarizing the metal fill layer 238 and completing the fabrication of the gate stack 230 as depicted in FIGS. 17A and 17B. In the present embodiments, the metal fill layer 238 and the portions of the WFM layers 236A-236D together constitute the top portion 230A, which is formed over the bottom portion 230B that includes portions of the WFM layers 236A-236D but is free of the metal fill layer 238.

Now turning back to FIG. 6A, at operation 422, the method 400 proceeds to forming the S/D contacts 220A-220E to interconnect various device-level features, such as the n-type S/D features 214N or the p-type S/D features 214P, with the subsequently-formed MLI structures. In the present embodiments, referring to FIGS. 5, 18A, and 18B, the S/D contacts 220A-220E are formed in the ILD layers 18 and 19 (and the ESL 250), and may include any suitable conductive material, such as Cu, W, Ru, Co, Al, other suitable materials, or combinations thereof. Each of the S/D contacts 220A-220E may further include a barrier layer comprising Ti, Ta, TiN, TaN, other suitable materials, or combinations thereof. Forming the S/D contacts 220A-220E may include forming the ILD layer 19 over the ILD layer 18, where the ILD layer 19 is substantially similar to the ILD layer 20 as discussed above, forming the ESL 250 over the ILD layer 19, forming contact openings (not depicted) in the ESL 250, the ILD layer 18, and the ILD layer 19 via a series of patterning and etching processes similar to those discussed above with respect to forming the p-type fins 206 and n-type fins 208, forming the silicide layer 223 over each of S/D features, i.e., the n-type S/D features 214N and the p-type S/D features 214P, and subsequently depositing the conductive material(s) over the silicide layer 223 in the contact openings by PVD, CVD, ALD, plating, other suitable deposition processes, or combinations thereof. The silicide layer 223 may be formed by first depositing a metal layer over each of the S/D features, performing a silicidation process to allow the metal layer to react with the materials of the S/D features, and subsequently removing un-reacted portions of the metal layer to leave behind the silicide layer 223. The silicide layer 223 may include nickel silicide, titanium silicide, cobalt silicide, other suitable silicides, or combinations thereof. Thereafter, any excess conductive material(s) may be removed by a CMP process to planarize a top surface of the S/D contacts 220A-220E.

At operation 424, referring to FIGS. 19A and 19B, the method 400 performs additional processing steps including, for example, separating or cutting the gate stack 230 into the gate stacks 231 and 232 as discussed above in reference to FIG. 4 and subsequently forming the MLI structure over the device 200. Cutting the gate stack 230 into two or more shorter gate stacks, e.g., the gate stacks 231 and 232, and inserting the gate isolation features 252 therebetween may include patterning to form trenches (not depicted) in the gate stack 230 where the gate isolation features 252 are desired, filling the trenches with a suitable dielectric material, and

subsequently planarizing the top surface of the gate isolation features 252 by one or more CMP process. The gate isolation features 252 may include silicon nitride, silicon carbide, oxygen-doped silicon nitride (SiON), carbon-doped silicon nitride (SiCN), aluminum nitride, other suitable materials, or combinations thereof. In some embodiments, the gate isolation features 252 and the ESL 250 are configured with different compositions. In the depicted embodiments, the gate isolation features 252 are formed at boundaries between adjacent NS CMOSFETs, i.e., between the NS NFET 202A and the NS NFET 202B. The gate isolation features 252 may extend to contact the top surface of the isolation structures 14 as depicted in FIG. 19B. Alternatively, the gate isolation features 252 may extend to below the top surface of the isolation structures 14.

The MLI structure may include metal layers, dielectric layers, and interconnect structures, such as the gate contacts 222A and 222B, configured to electrically couple various components (e.g., gate stacks, S/D features, etc.) of the device 200 with additional devices and components to ensure the proper performance of the IC structure 100. Forming vertical interconnect structures may include forming a dielectric layer, such as the ILD layer 20, over the gate stacks 231 and 232 and subsequently patterning the ILD layer 20 to form contact openings similar to those formed for the S/D contacts 220A-220E discussed above. Of course, additional vertical interconnect structures, such as vias 226A and 226B discussed above in reference to FIGS. 2, 4, and 5, as well as horizontal interconnect structures, metal layers, and dielectric layers may also be formed in the MLI structure over the device 200 by any suitable methods.

Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, the present embodiments provide a semiconductor structure including a metal gate stack engaged with a stack of channel layers to form an NS FET, where a top portion of the metal gate stack disposed over the stack of channel layers differs from a bottom portion of the metal gate stack interleaved with the stack of channel layers in composition. In the present embodiments, the top portion of the metal gate stack includes a metal fill layer disposed over a WFM layer, where sidewalls of the metal gate stack are continuous with sidewalls of the WFM layer, and the bottom portion of the metal gate stack includes the WFM layer filling space between the channel layer but is free of the metal fill layer. In some embodiments, the semiconductor structure includes two NS FETs disposed adjacent to each other, where the metal gate stacks configured to form the NS FETs differ in the composition of the WFM layer but share the same metal fill layer. Embodiments of the present disclosure provide methods of forming metal gate stacks in NS FETs with improved ability to tune V_{th} , lowered resistance of the metal fill layer, and greater design freedom in choosing gate contact locations for routing efficiency consideration.

In one aspect, the present embodiments provide a semiconductor structure that includes a substrate, a semiconductor fin protruding from the substrate, where the semiconductor fin includes semiconductor layers stacked in a vertical direction, a gate stack engaging with channel regions of the semiconductor fin, and S/D features disposed adjacent to the gate stack in S/D regions of the semiconductor fin. In the present embodiments, the gate stack includes a first portion disposed over the semiconductor layers and a second portion disposed between the semiconductor layers, where the first portion includes a WFM layer

and a metal fill layer disposed over the WFM layer and the second portion includes the WFM layer but is free of the metal fill layer.

In another aspect, the present embodiments provide a semiconductor structure that includes a stack of semiconductor layers disposed over a substrate, a gate structure disposed over the stack of semiconductor layers, and S/D features disposed in the stack of semiconductor layers, such that the gate structure is interposed between the S/D features. In the present embodiments, the gate structure includes a first metal layer and a second metal layer disposed over a top surface of the first metal layer, where the first metal layer fills in space between the stack of semiconductor layers and the first metal layer includes at least one WFM layer and the second metal layer differs from the first metal layer in composition.

In yet another aspect, the present embodiments provide a method that includes forming a fin including a multi-layer stack (ML) of alternating first semiconductor layers and second semiconductor layers over a substrate, where the first semiconductor layers and the second semiconductor layers differ in composition, forming a dummy gate stack over a channel region of the fin, forming S/D features in the fin, such that the dummy gate stack is interposed between the S/D features, removing the dummy gate stack to form a first trench, removing the second semiconductor layers to form openings between the first semiconductor layers, and subsequently forming a metal gate stack. In the present embodiments, forming the metal gate stack includes forming a first metal layer over the substrate, thereby filling the first trench to form a top gate stack and filling the openings to form a bottom gate stack, removing a top portion of the top gate stack to form a second trench, and forming a second metal layer in the second trench, where the second metal layer differs from the first metal layer in composition.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:

- a protrusion extending from a substrate;
- a plurality of semiconductor layers stacked in a vertical direction above the protrusion;
- a gate stack engaging with channel regions of the semiconductor layers, wherein the gate stack includes a first portion disposed over a top semiconductor layer of the semiconductor layers and a second portion disposed below the top semiconductor layer of the semiconductor layers, wherein the first portion includes a work-function metal (WFM) layer over the semiconductor layers and a conductive layer disposed over the WFM layer, and wherein the second portion includes the WFM layer, wherein a sidewall of the WFM layer aligns with a sidewall of the conductive layer; and
- source/drain (S/D) features disposed adjacent to the gate stack.

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2. The semiconductor structure of claim 1, wherein the WFM layer includes TiN, TaN, TiAl, TiAlN, TaAl, TaAlN, TaAlC, TaCN, WNC, or combinations thereof, and the conductive layer includes W, Co, Ru, or combinations thereof.

3. The semiconductor structure of claim 1, wherein the conductive layer has a first length spanning a length of the gate stack along a first direction, the WFM layer has a second length along the first direction, wherein the first length is greater than the second length.

4. The semiconductor structure of claim 1, further comprising a gate contact feature electrically coupled to the gate stack, wherein the gate contact feature is vertically offset from the semiconductor layers.

5. The semiconductor structure of claim 4, wherein the gate contact feature directly lands on the conductive layer.

6. The semiconductor structure of claim 1, wherein the conductive layer has less resistance than the WFM layer.

7. The semiconductor structure of claim 1, wherein a first thickness of the conductive layer in the first portion is at least the same as a second thickness of the WFM layer in the first portion, the first thickness and the second thickness being measured along the vertical direction.

8. The semiconductor structure of claim 1, wherein the second portion is free of the conductive layer.

9. The semiconductor structure of claim 1, further comprising:

a shallow trench isolation (STI) structure adjacent the protrusion, wherein a top surface of the STI structure is lower than a top surface of the protrusion.

10. A semiconductor structure, comprising:

a stack of semiconductor layers disposed over a substrate; a gate structure disposed over the stack of semiconductor layers,

wherein the gate structure includes a gate dielectric layer wrapping around each semiconductor layer in the stack and over a topmost semiconductor layer of the stack, a first conductive layer over the gate dielectric layer and filling in spaces among the semiconductor layers in the stack and having a portion above the topmost semiconductor layer of the stack, and a second conductive layer disposed on the first conductive layer, wherein the first conductive layer includes at least one work-function metal (WFM) layer, and the second conductive layer differs from the first conductive layer in composition, wherein a top surface of the gate dielectric layer is above a top surface of the first conductive layer; and a gate contact structure landing on the second conductive layer.

11. The semiconductor structure of claim 10, wherein the stack of semiconductor layers is a first stack of semiconductor layers, the gate structure is a first gate structure, and the gate dielectric layer is a first gate dielectric layer, the semiconductor structure further comprising:

a second stack of semiconductor layers disposed over the substrate and adjacent to the first stack of semiconductor layers; and

a second gate structure disposed over the second stack of semiconductor layers,

wherein the second gate structure includes a second gate dielectric layer wrapping around each semiconductor layer in the second stack and over a topmost semicon-

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ductor layer of the second stack, a third conductive layer over the gate dielectric layer and filling in spaces among the semiconductor layers in the second stack, and the second conductive layer disposed over a top surface of the third conductive layer, wherein the third conductive layer includes at least one WFM layer, and the third conductive layer differs from the first conductive layer in composition.

12. The semiconductor structure of claim 11, further comprising:

n-type source/drain features adjacent to the semiconductor layers in the first stack; and

p-type source/drain features adjacent to the semiconductor layers in the second stack.

13. The semiconductor structure of claim 10, wherein the second conductive layer has a sidewall that directly contacts a sidewall of the gate dielectric layer.

14. The semiconductor structure of claim 10, wherein gate contact structure is vertically aligned with the stack of semiconductor layers.

15. The semiconductor structure of claim 10, wherein gate contact structure is vertically offset from the stack of semiconductor layers.

16. The semiconductor structure of claim 10, wherein the first conductive layer includes one or more WFM selected from TiN, TaN, TiAl, TiAlN, TaAl, TaAlN, TaAlC, TaCN, WNC, or combinations thereof, wherein the second conductive layer includes W, Co, Ru, or combinations thereof.

17. A semiconductor structure, comprising:

a substrate having a protrusion;

a plurality of semiconductor layers stacked in a vertical direction above the protrusion; and

a gate stack engaging with channel regions of the semiconductor layers, wherein the gate stack includes a top portion over a bottom portion, wherein the bottom portion includes a gate dielectric layer wrapping around each of the semiconductor layers and a work-function metal (WFM) layer wrapping around the gate dielectric layer, and the top portion includes a conductive layer landing on the WFM layer and interfacing sidewall portions of the gate dielectric layer,

wherein a topmost surface of the WFM layer and a topmost surface of the conductive layer are on different levels,

wherein a top surface of the gate dielectric layer is coplanar with the topmost surface of the conductive layer.

18. The semiconductor structure of claim 17, wherein the conductive layer has a first thickness, the WFM layer has a second thickness, and the second thickness is greater than the first thickness.

19. The semiconductor structure of claim 18, wherein the conductive layer has less resistance than the WFM layer.

20. The semiconductor structure of claim 17, further comprising:

gate isolation features extending along sidewalls of the gate stack, wherein each of the top and the bottom portions of the gate stack spans an entire width between the gate isolation features, wherein the gate stack has a top surface below top surfaces of the gate isolation features.

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